

Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

THOMAS DUPIC

Which antigens are targeted by intra-tumoral B cells?



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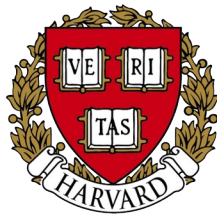
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Post-Doc
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T. Mora, Paris
*Statistics of the
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6 pub., 2 1st aut.

LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE



2018

2020

2025



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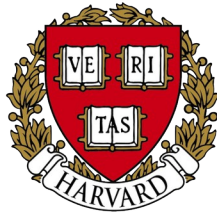
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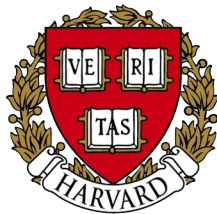
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Team Leader at CIML, Marseille
*Affinities within the immune
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Hybrid team, experimental and
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Ciml
IMMUNOLOGY

2025



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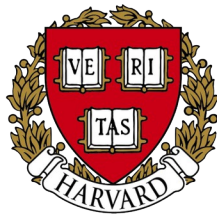


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CSS 2: Cancer &
Genetic diseases

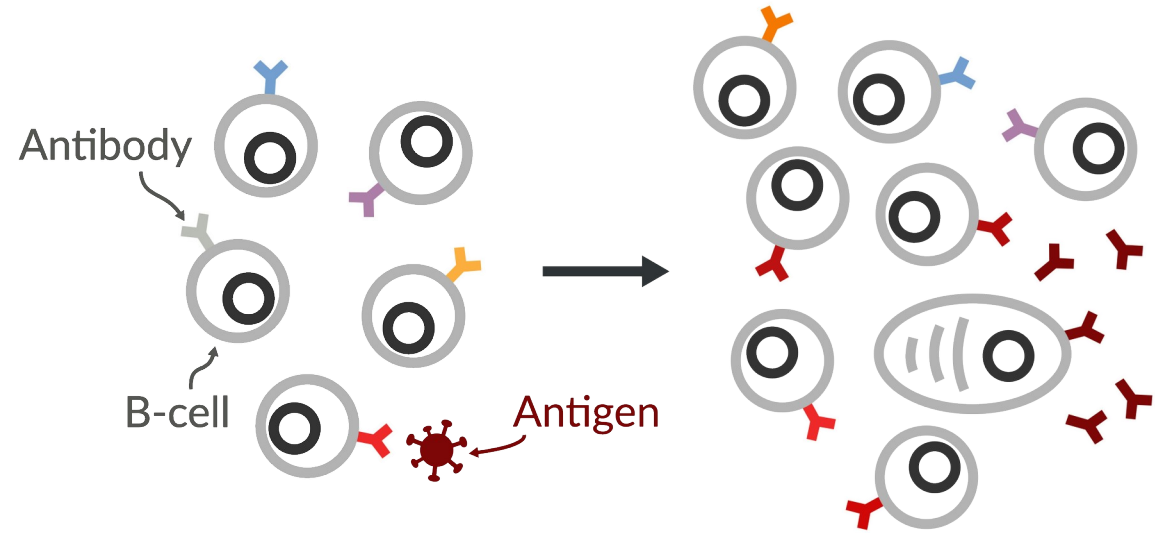
Inserm

Immune repertoires

Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).

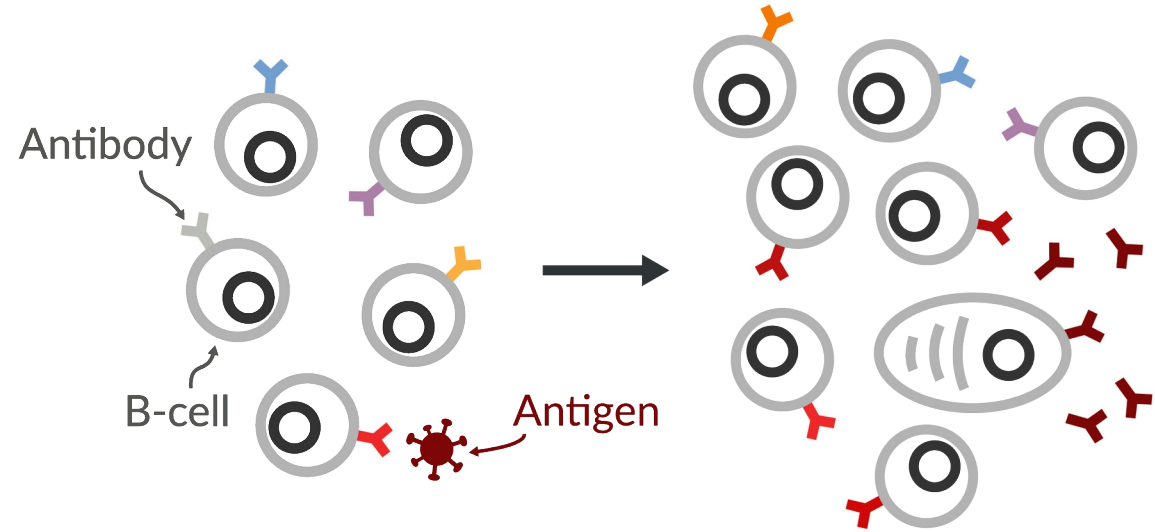
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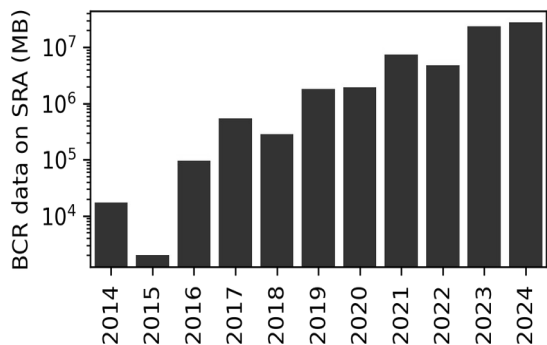
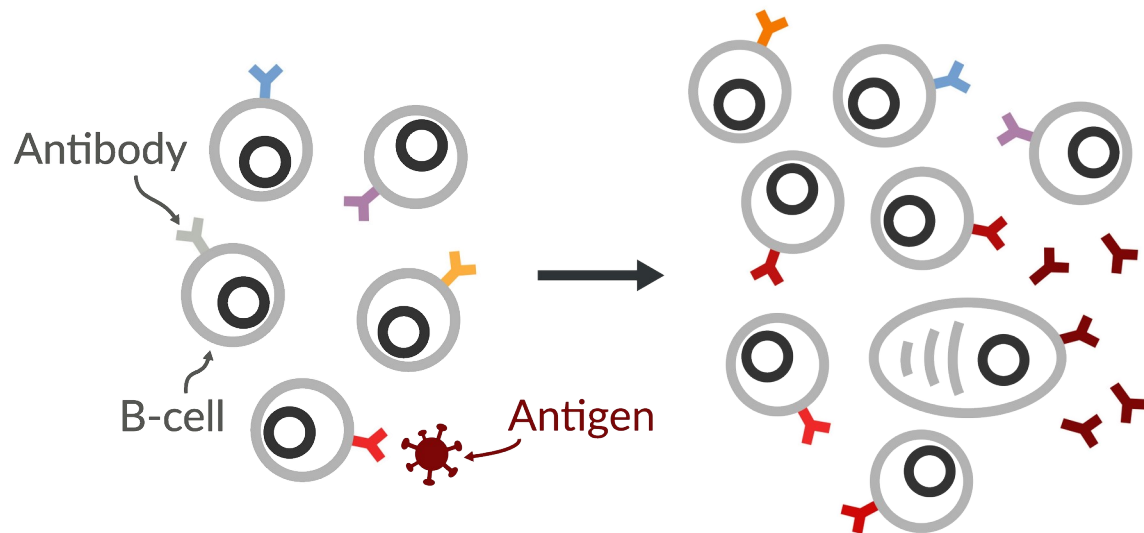
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The repertoire as a prognosis tool ?

Immune repertoires

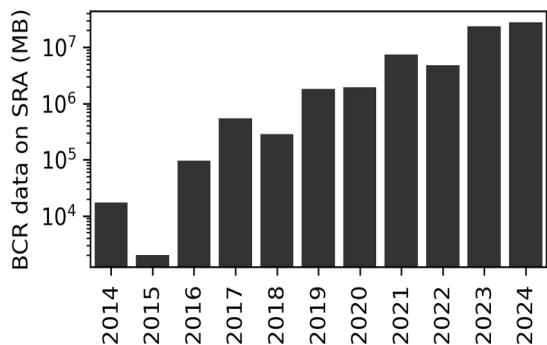
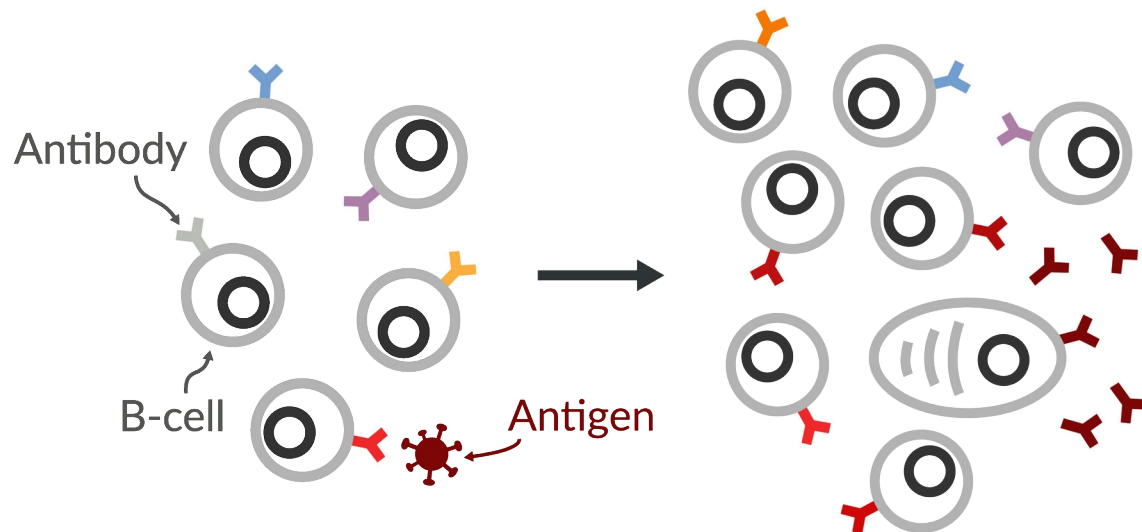
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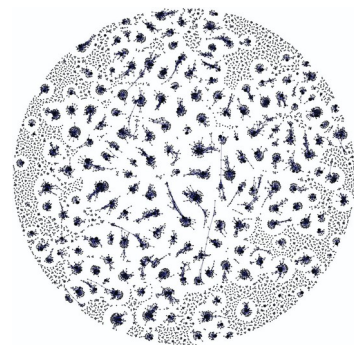
The repertoire as a prognosis tool ?
Easy to access...

Immune repertoires

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The repertoire as a prognosis tool ?
Easy to access... but hard to interpret.



Immune repertoires

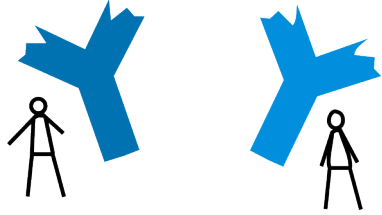
Statistical models, **generation**¹ & **selection**² of the receptors. **Dupic et al., PLOS Comp. Bio (2019)**

$$P(\text{Y}) = \dots$$

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Public repertoire: do shared receptors imply a shared response?

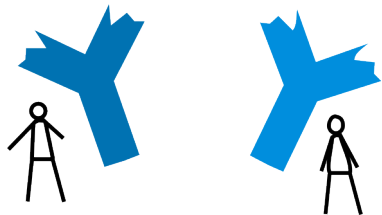
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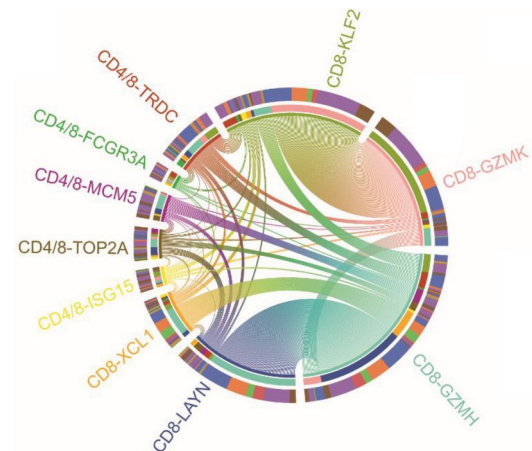
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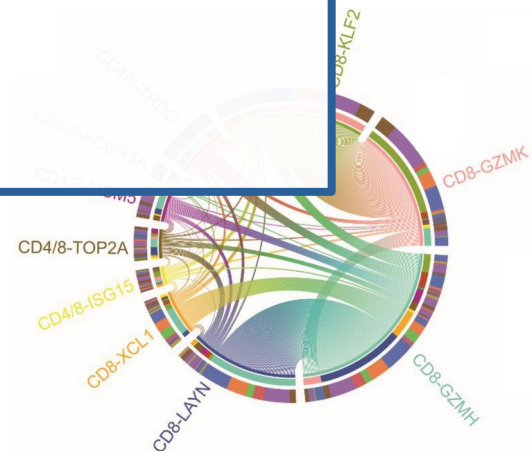
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shared Response?

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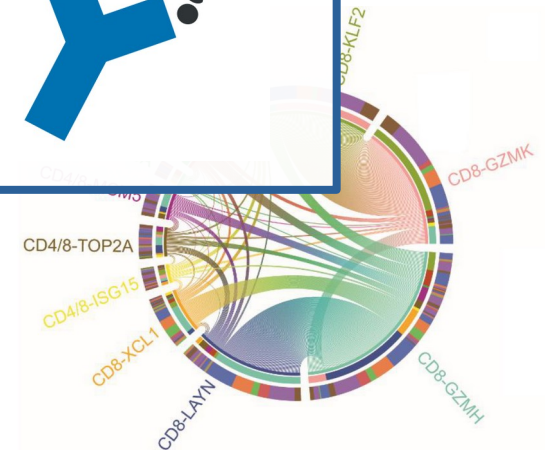
Current methods don't integrate the **functional properties** of the receptor.



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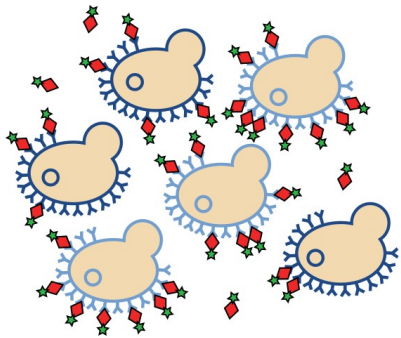
Beyond barcodes : tracking interactions between proteins in high-throughput

Tite-Seq: high-throughput and quantitative method can measure up to 100'000 proteins against one ligand.



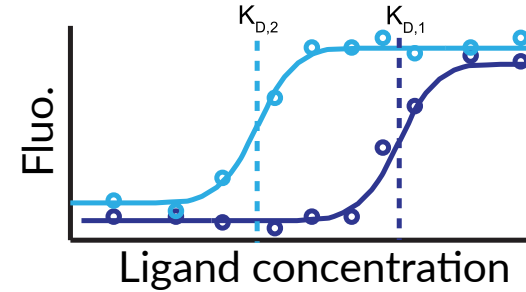
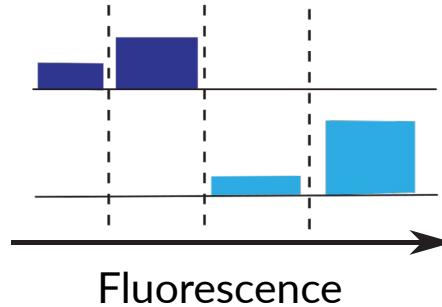
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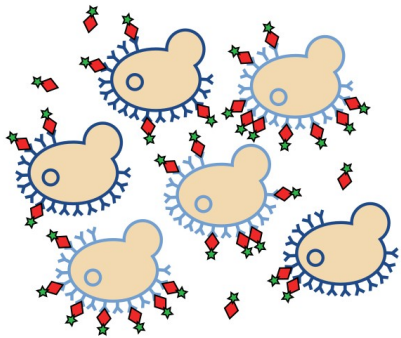
Expression on the yeast with fluorescent antigen

Cell sorting, sequencing and statistics



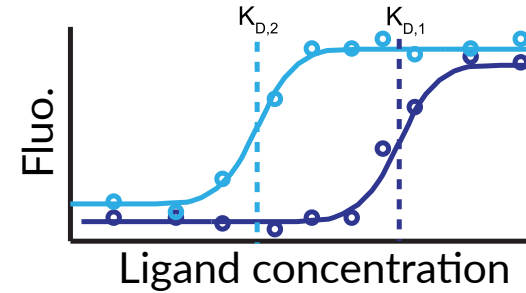
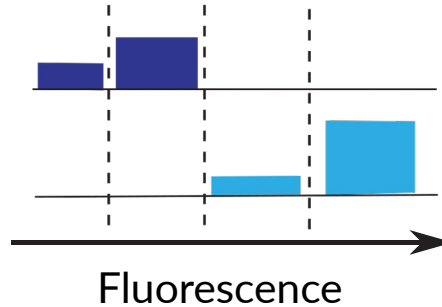
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Viral proteins

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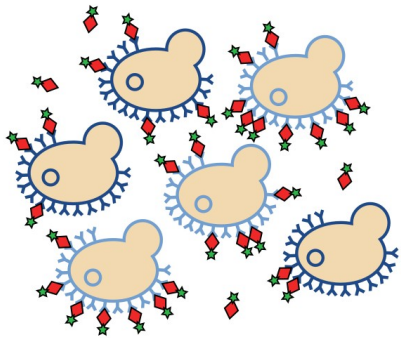
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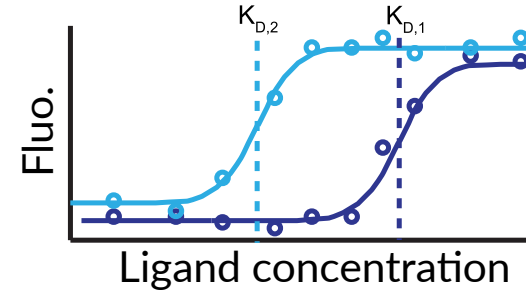
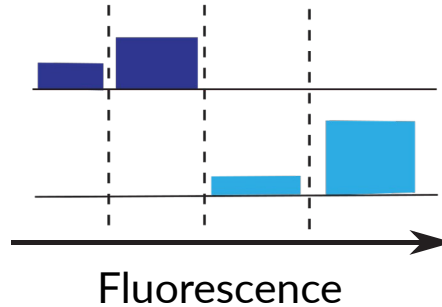
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Antibodies

Phillips, ..., Dupic et al., eLife (2021)

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Dupic, et al., Genetics (2026, in revision)



Measuring affinity at repertoire scale



Measuring affinity at repertoire scale

Long-term B-cell
response to SARS-
CoV-2 vaccination

Moulana, ..., Dupic*, Desai* (in preparation)



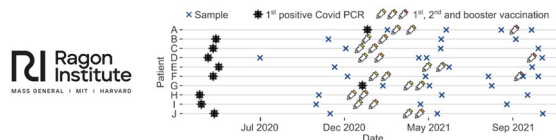
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Repertoire Sequencing →



Common BCRs →

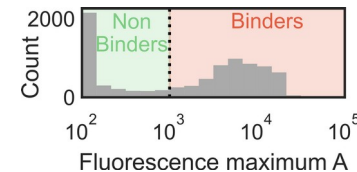


Affinity

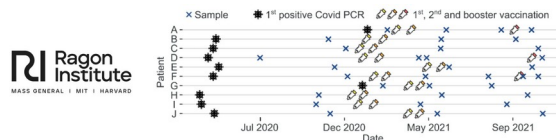
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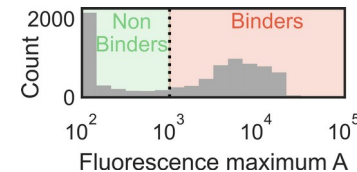
Affinity

1M BCRs tested $\Rightarrow$ 5000 binders

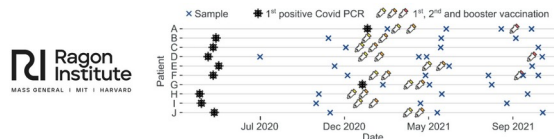
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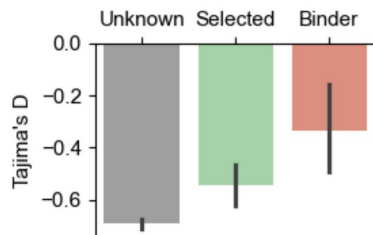
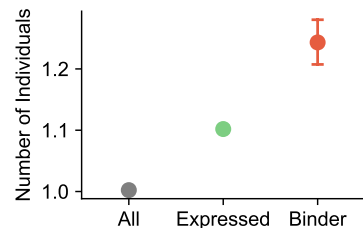
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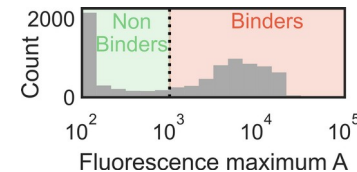


The repertoire statistics of **binders**
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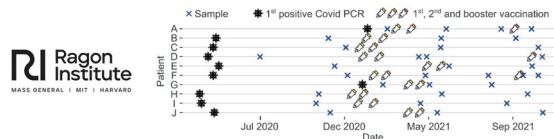
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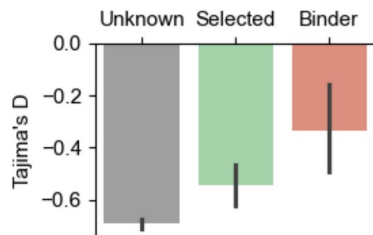
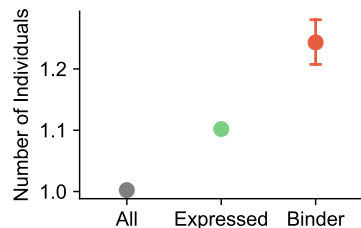
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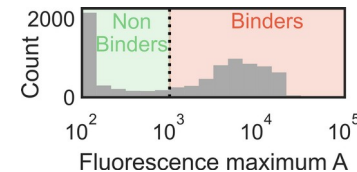
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Improving binding models with transfer learning

Moulana, ..., Duplic*, Desai* (in preparation)

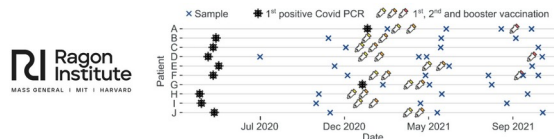


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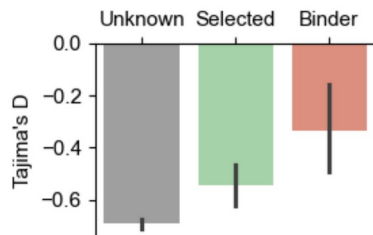
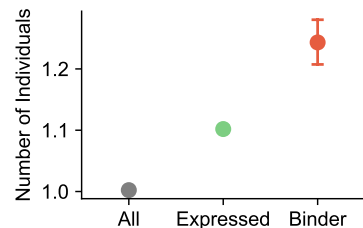
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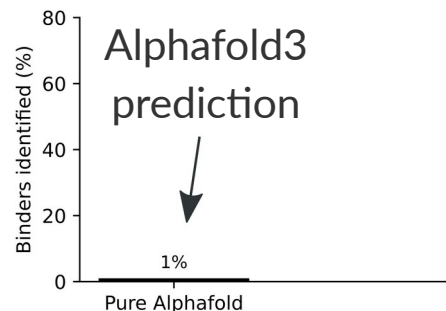
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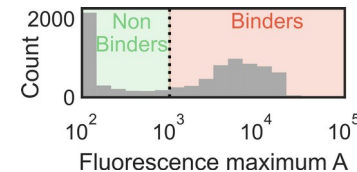
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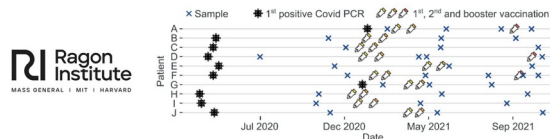


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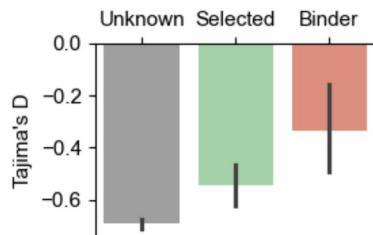
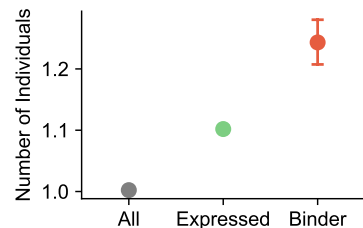
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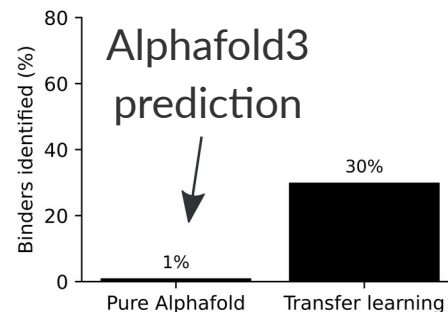
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AlphaFold3 embedding
 $\times$
Our measurements

Measuring affinity at repertoire scale

Tite-Seq × repertoire :



Sequenced
repertoires

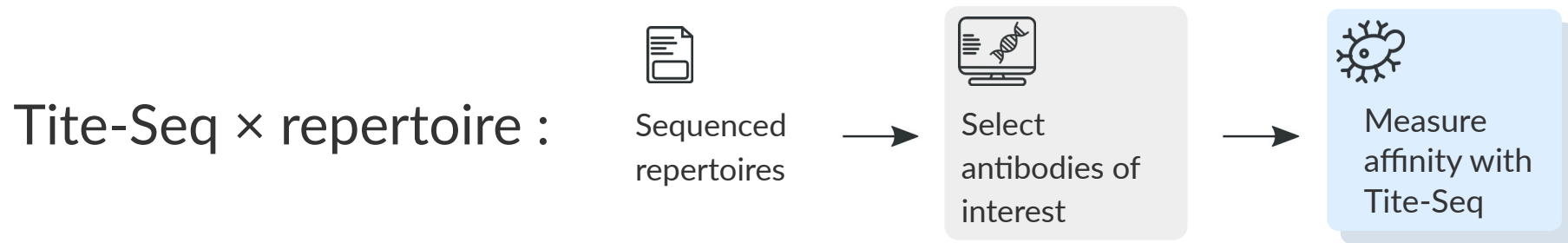


Select
antibodies of
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Measure
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Measuring affinity at repertoire scale



Tite-Seq is precise, mature and cheap

Measuring affinity at repertoire scale

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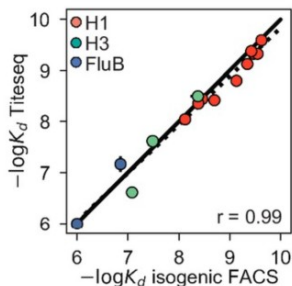


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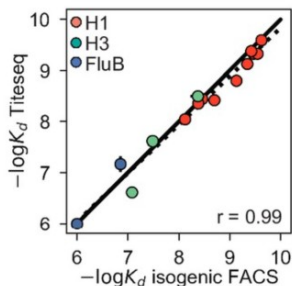


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Starr et al., Nature, 2021
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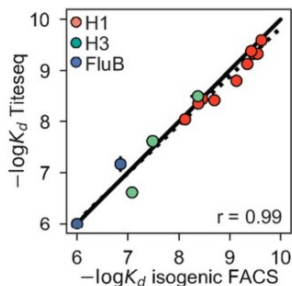


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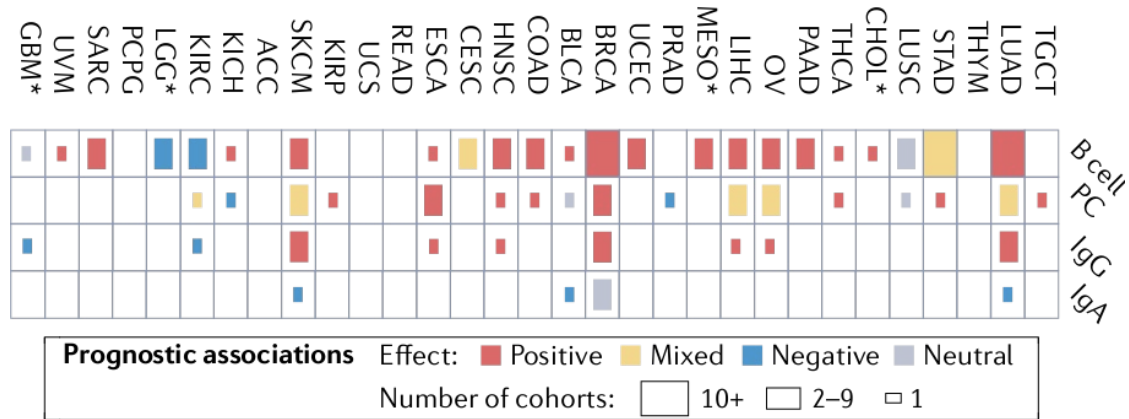


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Tumor Infiltrated B-Lymphocytes (TIL-B)

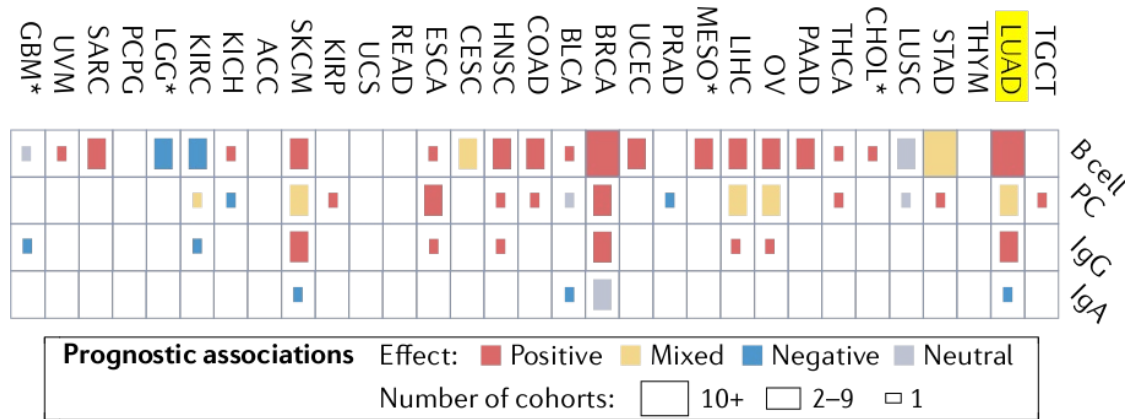
TIL-Bs play a complex role in tumors but generally improve prognosis and response to immunotherapy.



Adapted from *Laumont et al., Nat. Rev. Cancer (2022)*

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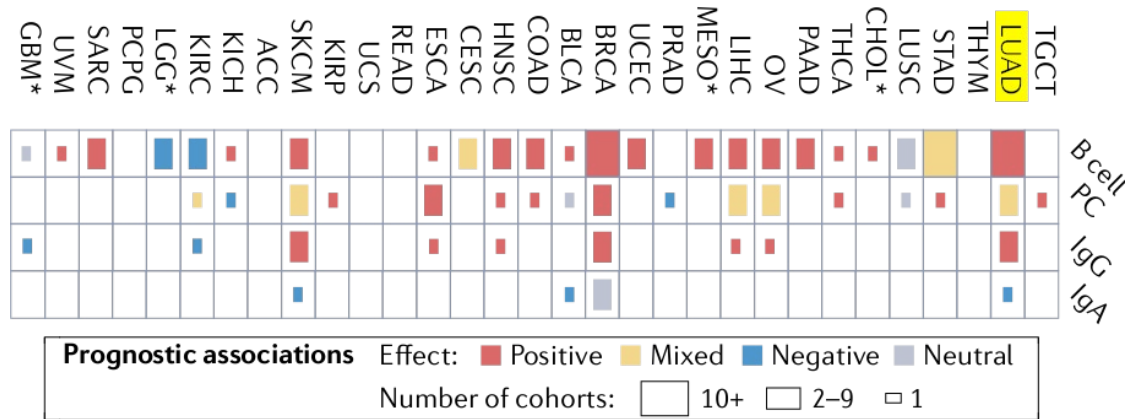
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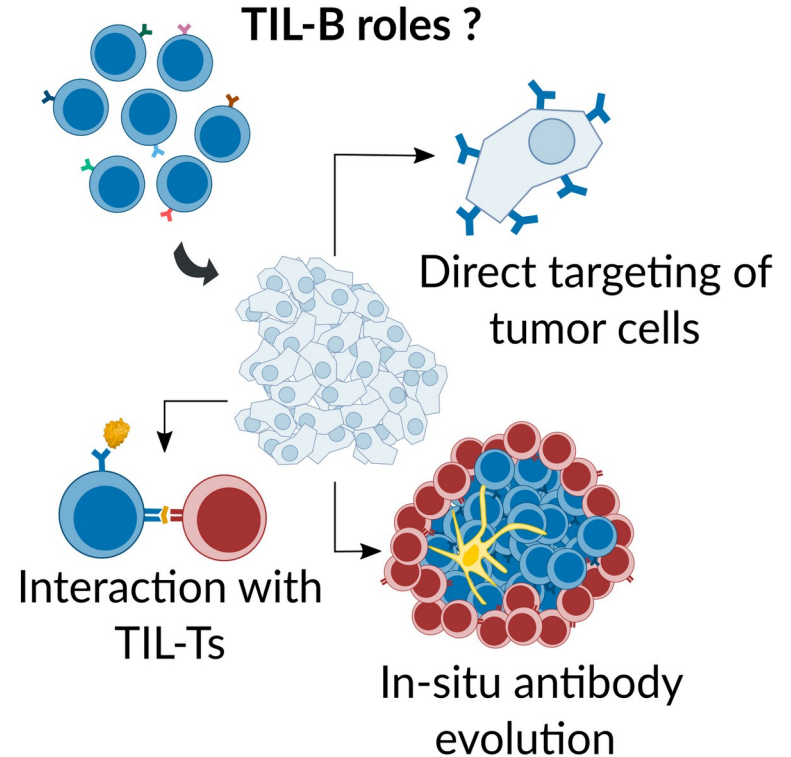
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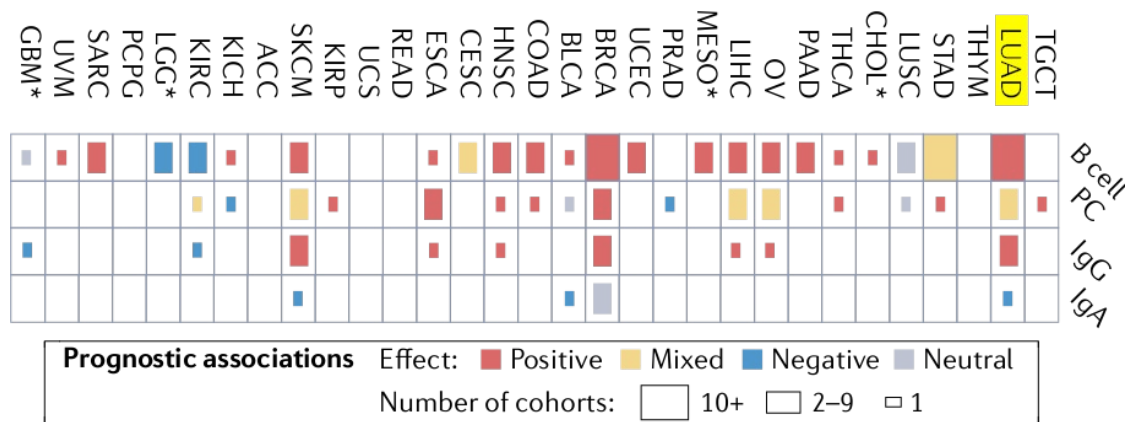


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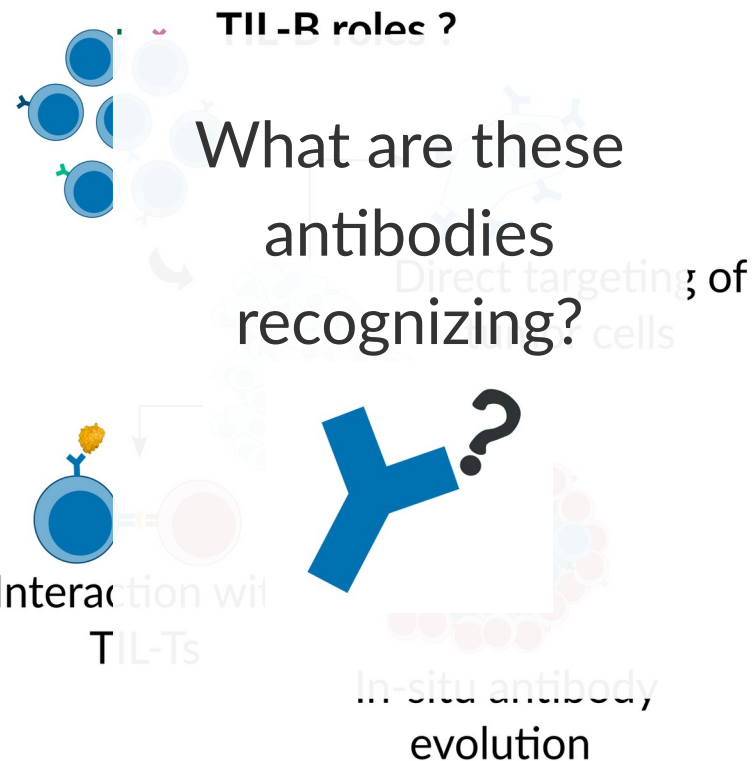


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The public TIL-B repertoire

Only **70 pairs** antigen /
TIL-antibody identified.

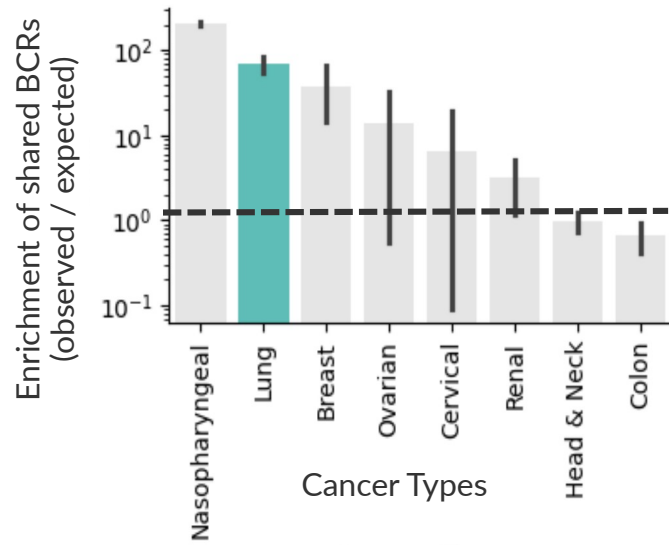
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The intra-tumoral repertoire is
public $\Rightarrow$ directed response !



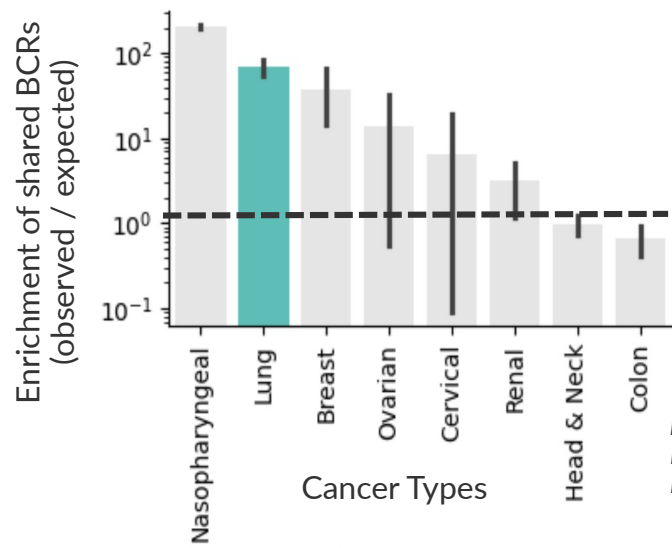
Data from: Aran et al, *Front. In Immunol.*, 2021, McDaniel et al., *Cancer Immunol Immunother*, 2018, Liu et al., *Nat. Commun.*, 2024, Shu et al., *BiorXiv*, 2023, Ma et al., *Science*, 2024, Ruffin et al., *Nat. Commun.*, 2024

The public TIL-B repertoire

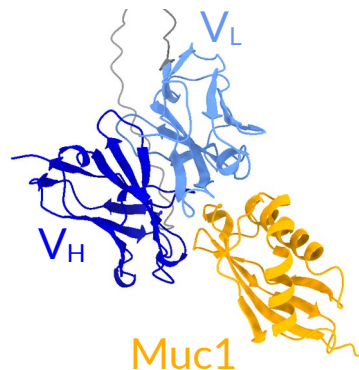
Only **70 pairs** antigen /
TIL-antibody identified.

Laumont et al., Nat. Rev. Cancer (2022)

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Preliminary Data

Lung cancer:

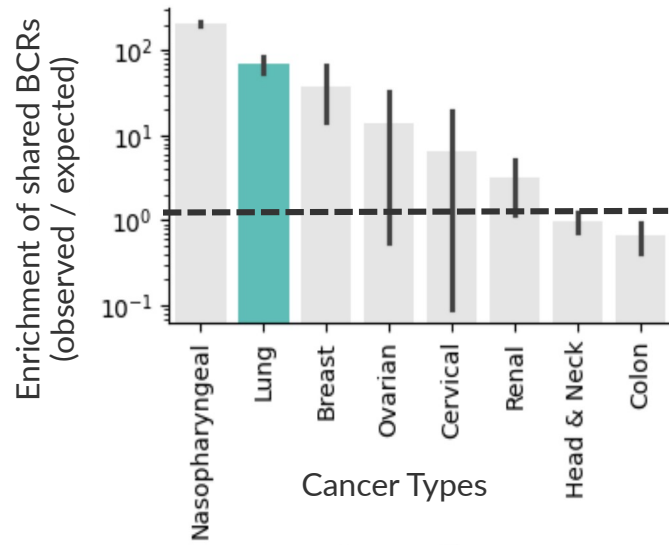
500 BCRs tested
in-silico against 5
antigens

The public TIL-B repertoire

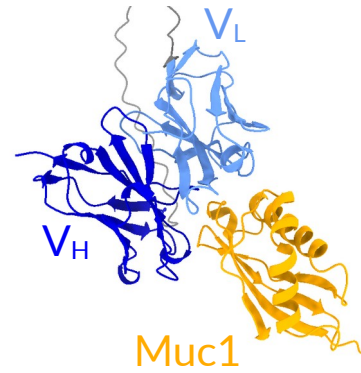
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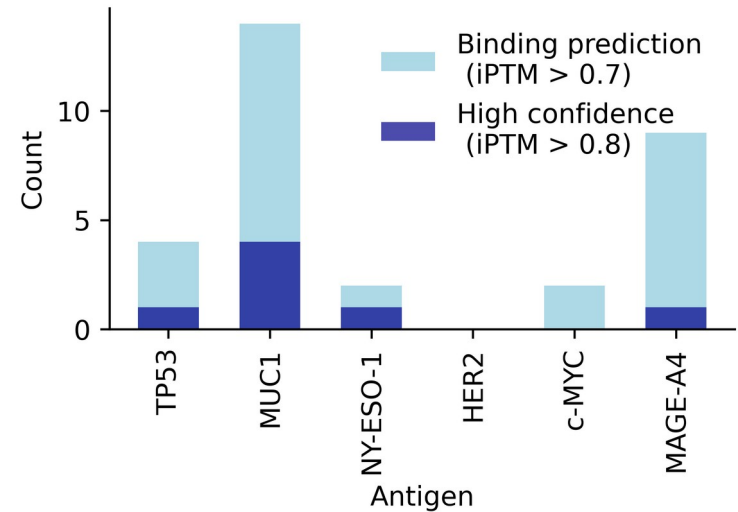


Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024



Preliminary Data

Lung cancer:
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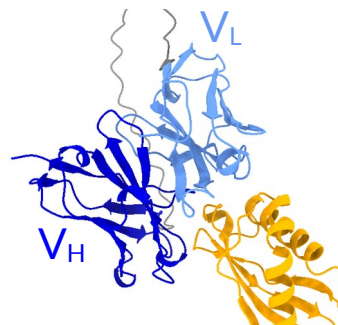
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Preliminary Data

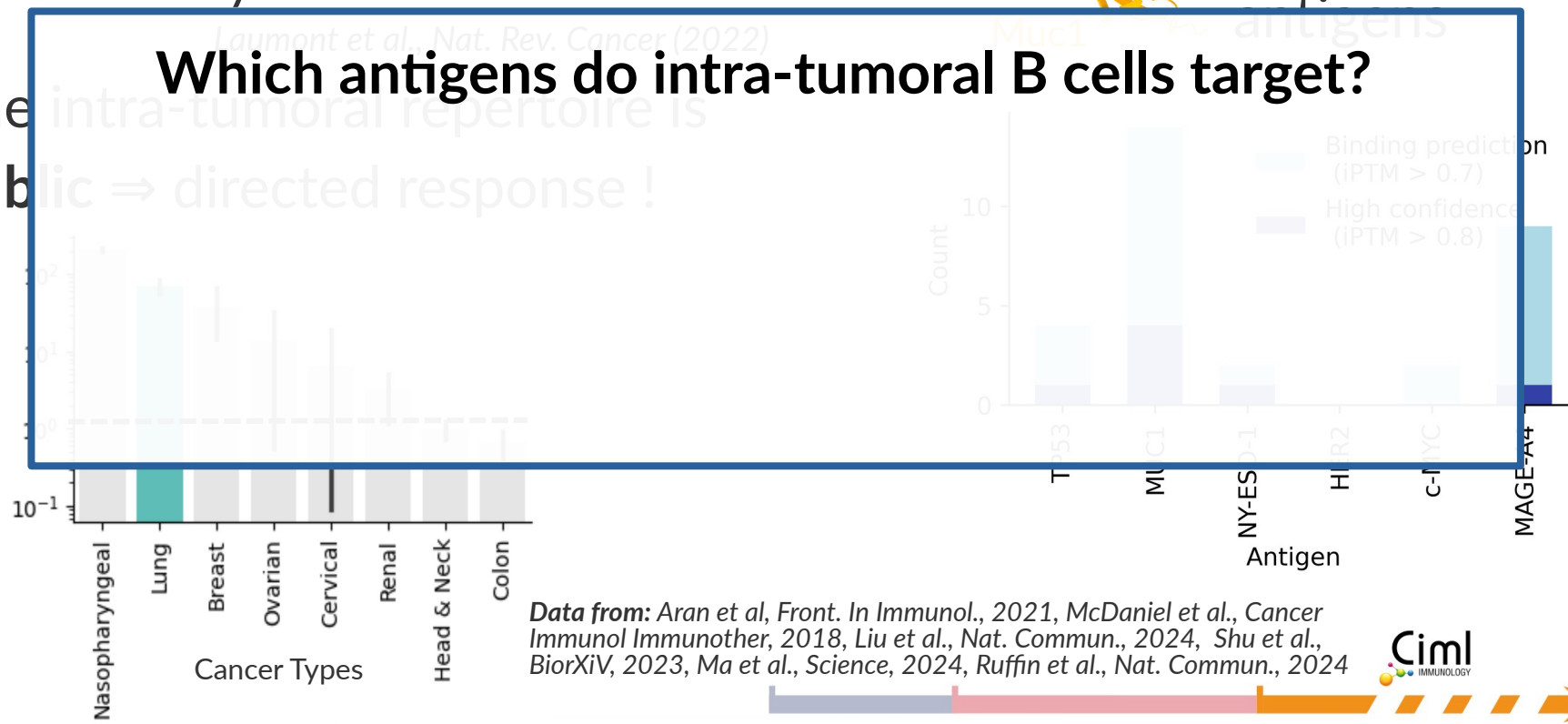
Lung cancer:

500 BCRs tested
in-silico against 5



Which antigens do intra-tumoral B cells target?

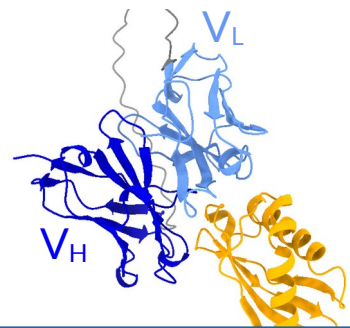
Enrichment of shared BCRs
(observed / expected)



Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

The public TIL-B repertoire

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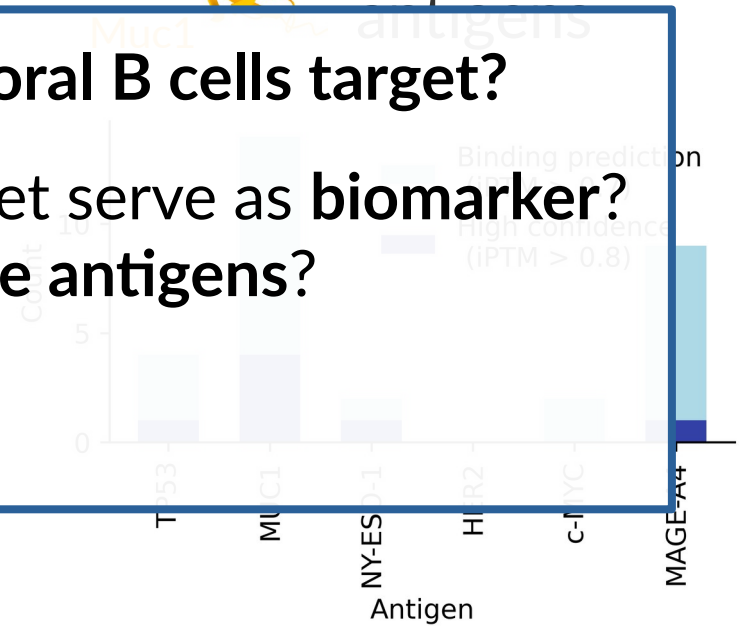
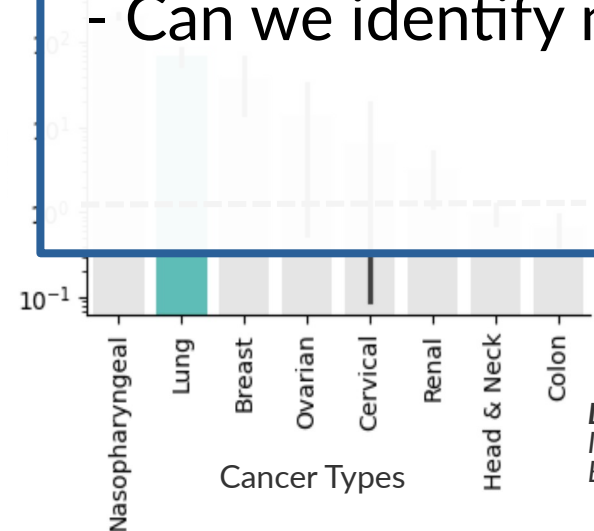


Lung cancer:
500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

- Can the antigens the BCRs target serve as **biomarker**?
- Can we identify novel **targetable** antigens?

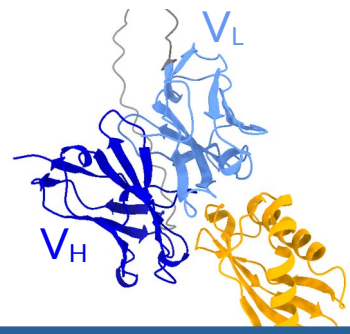
Enrichment of shared BCRs
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Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

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Only 70 pairs antigen /
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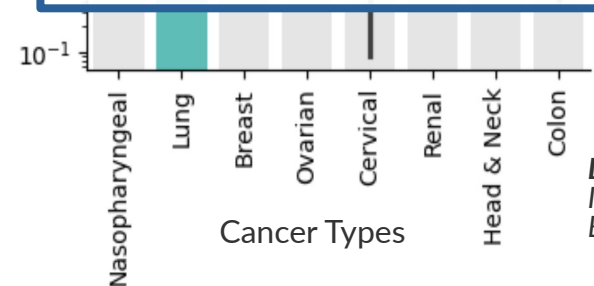
Lung cancer:

500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

- Can the antigens the BCRs target serve as **biomarker**?
- Can we identify novel **targetable** antigens?
- Do plasma cells have different specificity?
- Does specificity change during affinity maturation?

Enrichment of shared BCRs
(observed / expected)



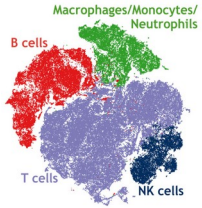
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Which antigens are targeted by TIL-Bs



Non-small cell lung
cancer dataset(s)



Public data

Hao (2022)

Prazanowska (2023)

...

RHU
LUCA-PI

900 prospective patients
(pulmonary nodules,
PBMC)



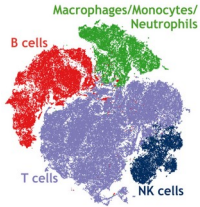
216 patients
(tumor & PBMC
samples)



Which antigens are targeted by TIL-Bs



Non-small cell lung cancer dataset(s)



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...

RHU
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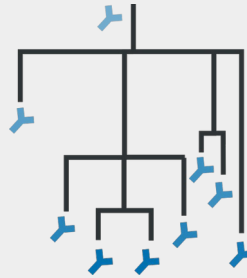
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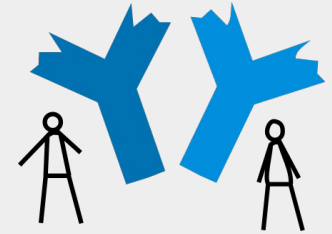


Identify TIL-B of interest *in-silico*
(WP2)



Clonality, isotypes &
transcriptomics

Plasma / Memory B-cells.
Clonal lineages under
selection



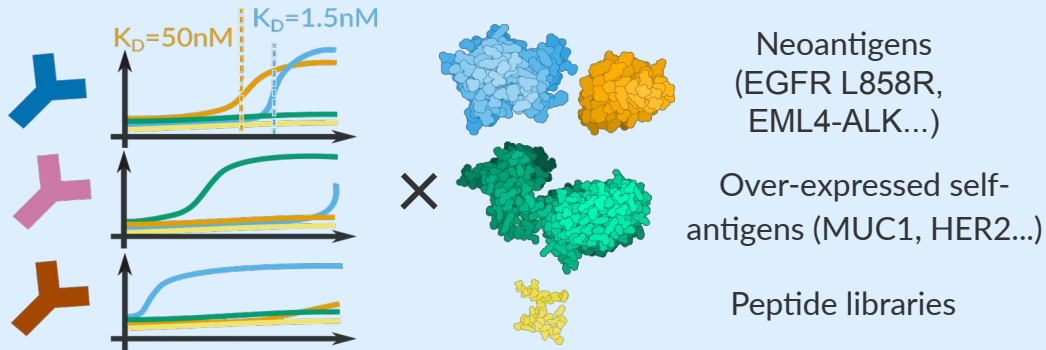
Comparisons
between patients

Identify similar receptors
between repertoires

Which antigens are targeted by TIL-Bs



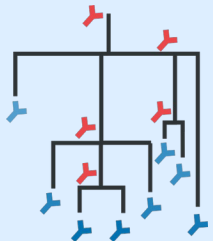
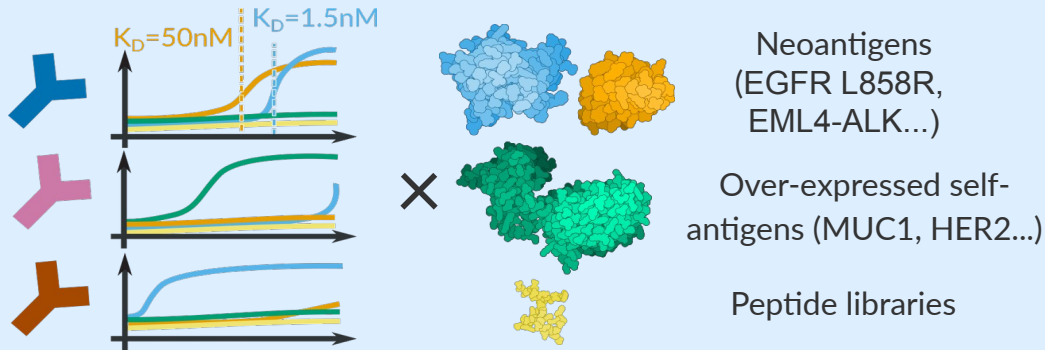
Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)

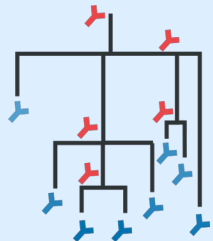
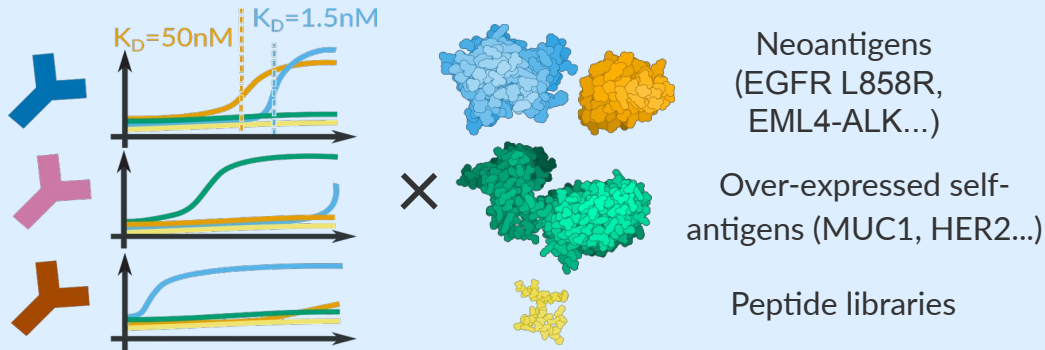


Ancestral
reconstruction
Did the antibody evolve
against this antigen ?

Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



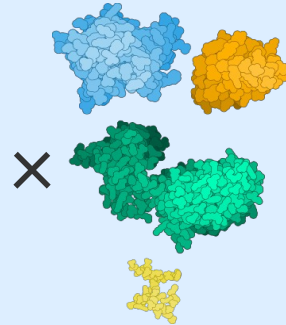
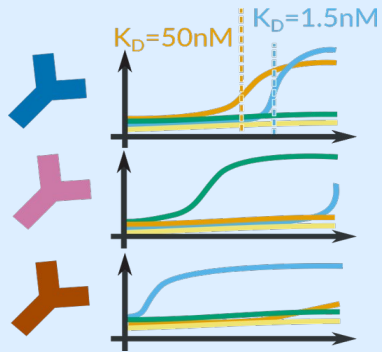
Ancestral reconstruction
Did the antibody evolve against this antigen ?

Comparison with sera / PBMC
Pioneer / LUCA-PI Cohort
PhiP-Seq

Which antigens are targeted by TIL-Bs



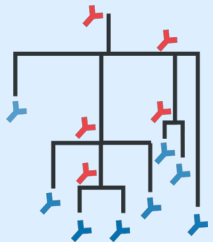
Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



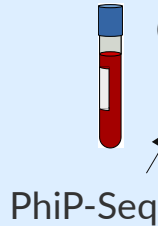
Neoantigens
(EGFR L858R,
EML4-ALK...)

Over-expressed self-
antigens (MUC1, HER2...)

Peptide libraries



Ancestral
reconstruction
Did the antibody evolve
against this antigen ?



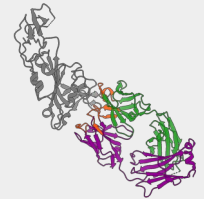
Comparison with
sera / PBMC

Pioneer / LUCA-
PI Cohort



Models &
learning (WP2)

Generalize Ab-Ag
predictions



Integrate specificity
to prognosis



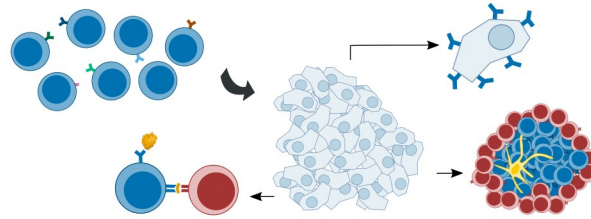
?

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer



Project summary



Which antigens do intra-tumoral B cells target?

- BCRs as biomarkers
- New actionable antigens ?
- Plasma cell specificity
- Evolution of specificity

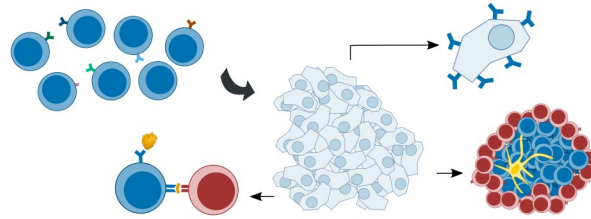


THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

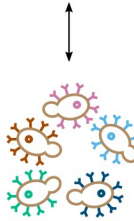


Project summary

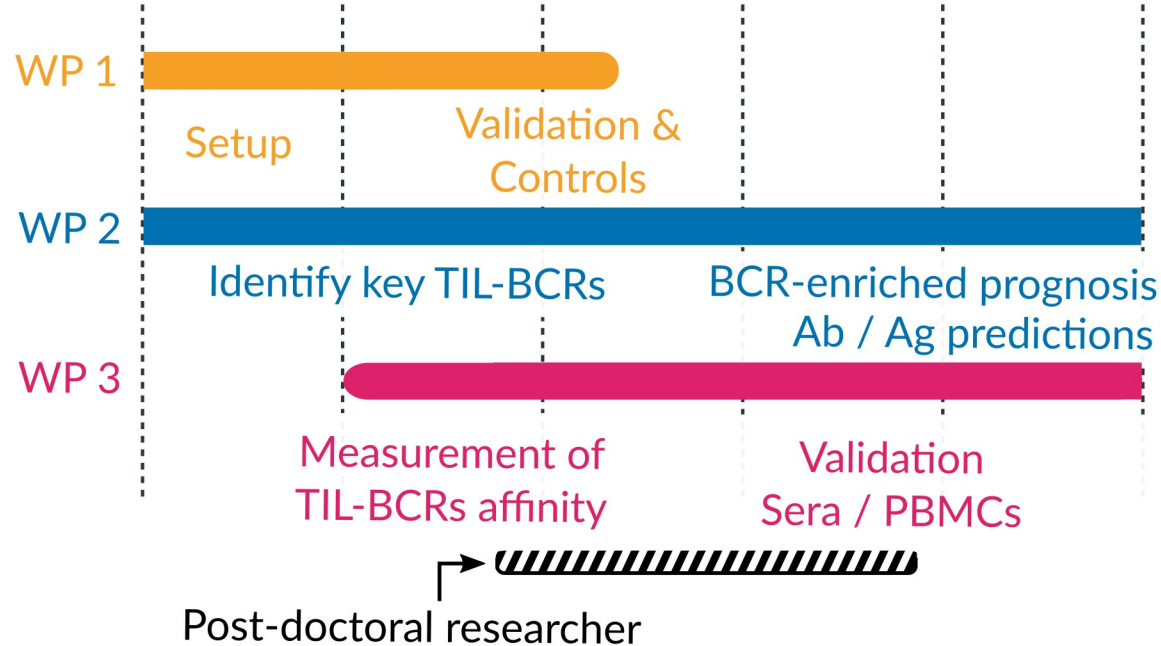


Which antigens do intra-tumoral B cells target?

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Timeline

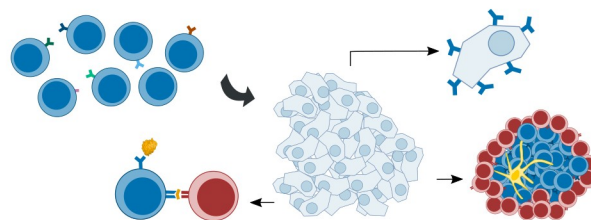


THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

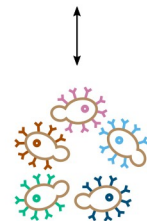


Project summary



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Partner institutions

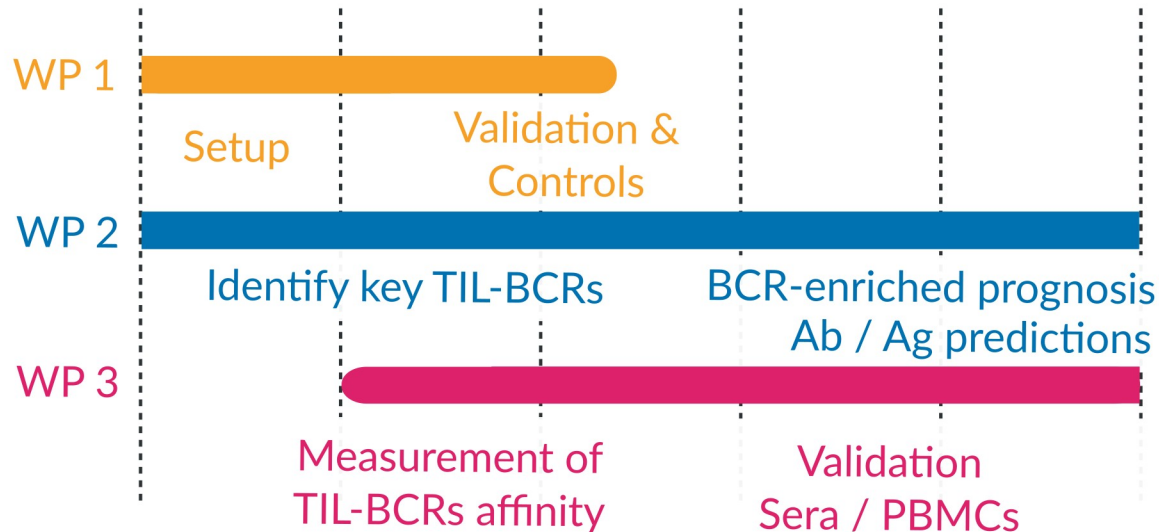


Co-financing
Idex $\Rightarrow$ 1 PhD



Team leader
Equipment,
lab, technician

Timeline



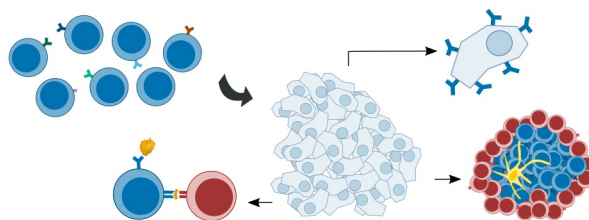
Post-doctoral researcher

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

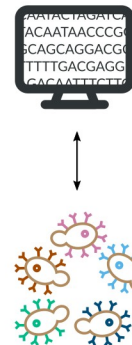


Project summary

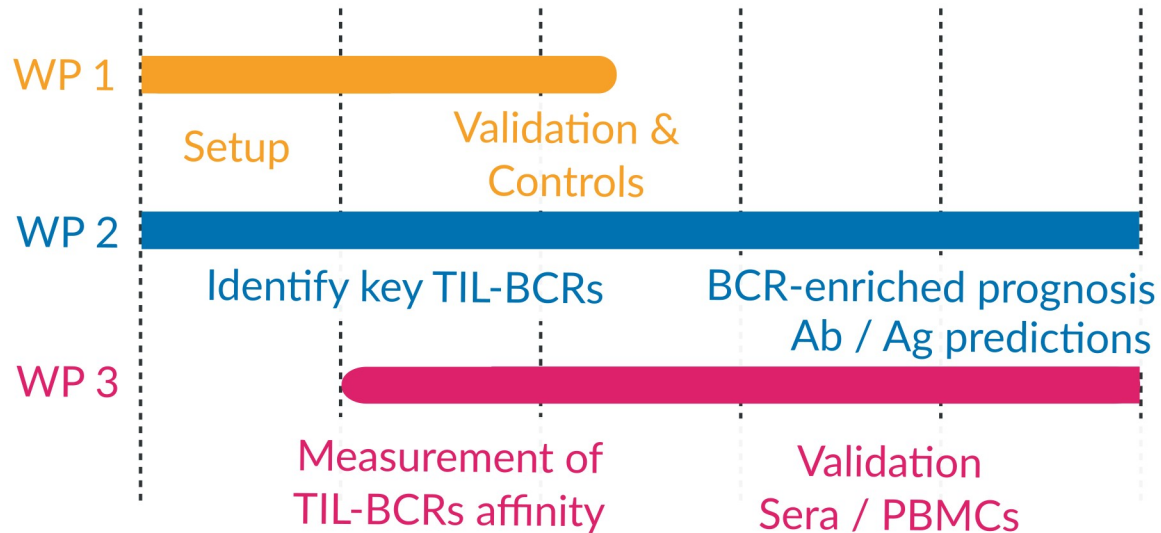


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Timeline



Post-doctoral researcher

Partner institutions

Inserm

amU
Aix Marseille Université

Co-financing
Idex $\Rightarrow$ 1 PhD

ap. hm

Cimi
IMMUNOLOGY

Team leader
Equipment,
lab, technician

Collaborations

CRCM
Centre de Recherche
en Cancérologie de Marseille

UCSF
 LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE

P. Milpied
Q. Marcou
A. Phillips
T. Mora, A. Walczak

LUCA-PI

THE PIONEER PROJECT

Additional slides

Atip-Avenir funding

Post-doctoral fellow recruitment ($\approx$ 2027 - 2029)

Budget estimates:

- DNA libraries ($\approx$ 50%) $\Rightarrow$ collections of antibody / antigens to be expressed
- Sequencing ($\approx$ 20%)
- General lab supplies ($\approx$ 20%)
- Conferences participation / paper costs ($\approx$ 10%)

The IDEX program of amU contributes with additional funding for one PhD student (2026 – 2029).

Pioneer project

Better understanding, predicting, and overcoming resistance to anti-PD-1(L1) therapies in lung cancer and beyond

Project finished in 2023.

BCR repertoire within
the tumor



Unused (not sequenced) blood
samples:

PBMC samples from 84 NSCLC
patients treated with PD-1(L1).

1 pre-treatment sample,

1-4 post-treatment samples.

225 total samples.

LUCA-Pi Project

Early prognostic of Lung Cancer

Project started in 2024

Nodule screening via imaging and B/T-cell repertoire sequencing in blood and nodules.



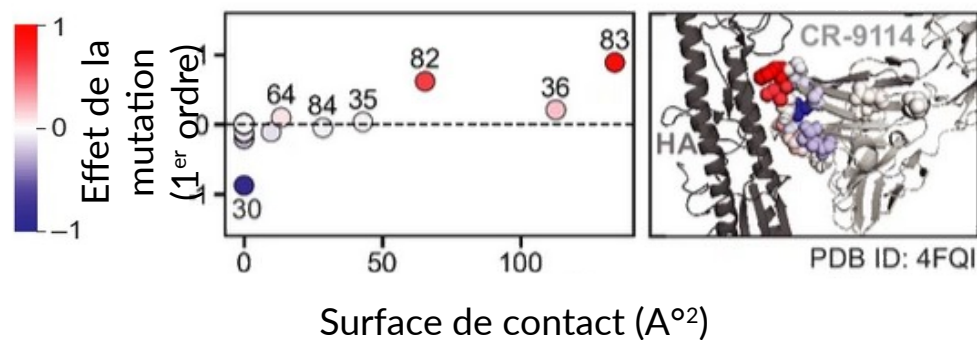
900 samples from at-risk individuals (heavy smokers), blood and pulmonary nodule.

Estimate: 10-15% of cancerous nodules.

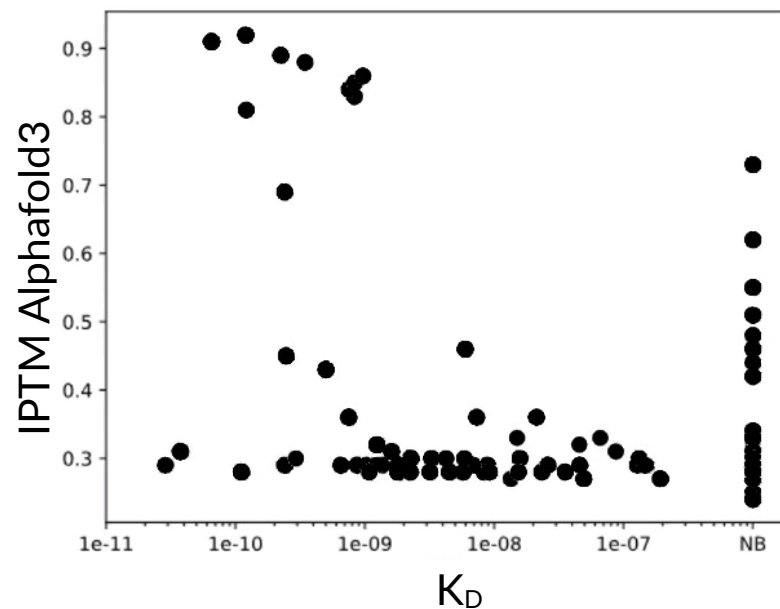
AI

AlphaFold3 / ProtBERT / AbMAP ...

Structure $\neq$ Affinity



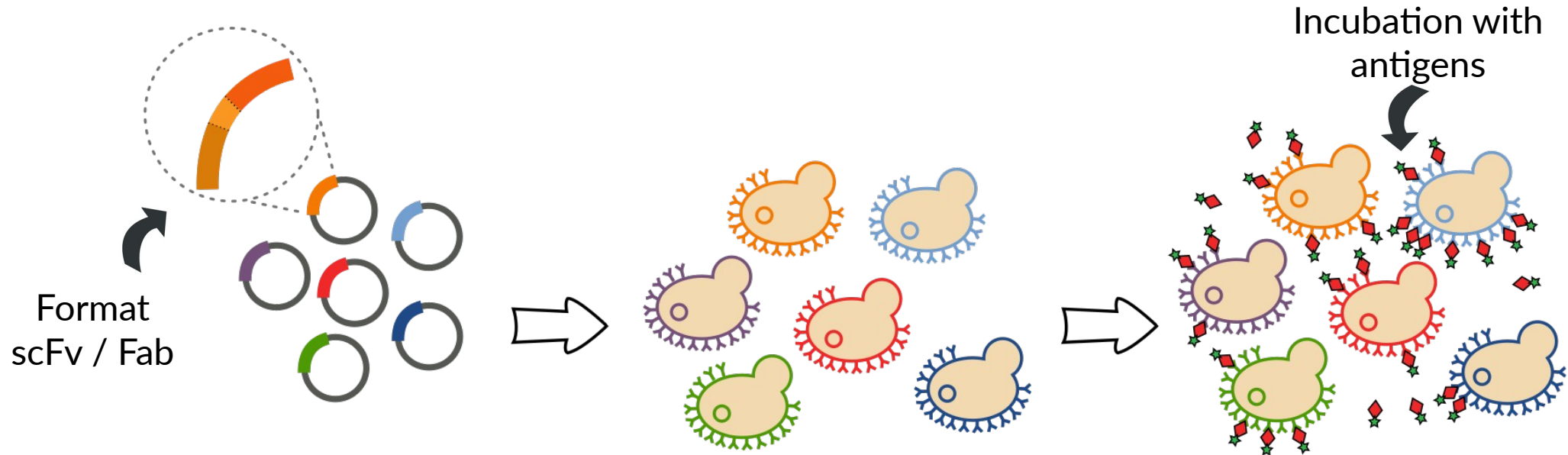
Antibodies that are not very specific are really a challenge for current methods



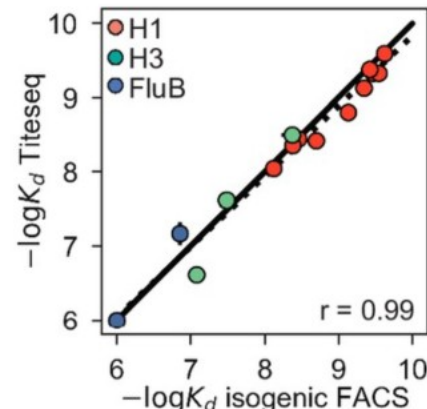
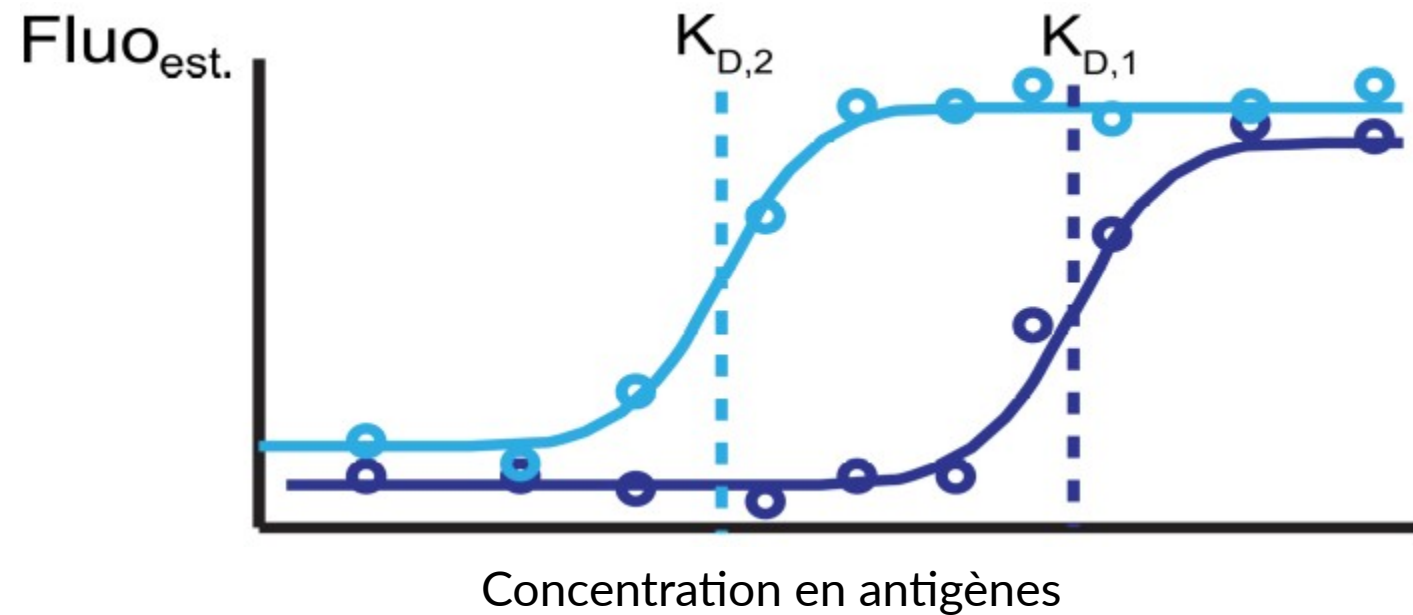
SARS-CoV-2 antibodies,
AlphaFold3 vs K_D

Tite-Seq

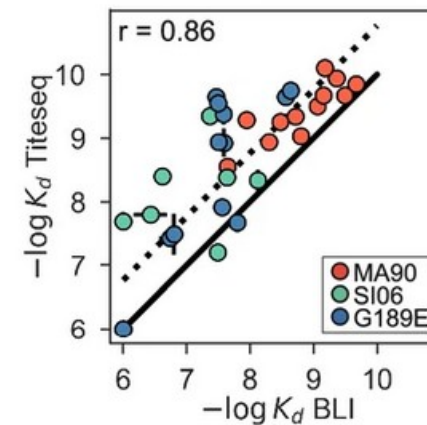
Antibodies expression (scFv) on yeast



Tite-Seq

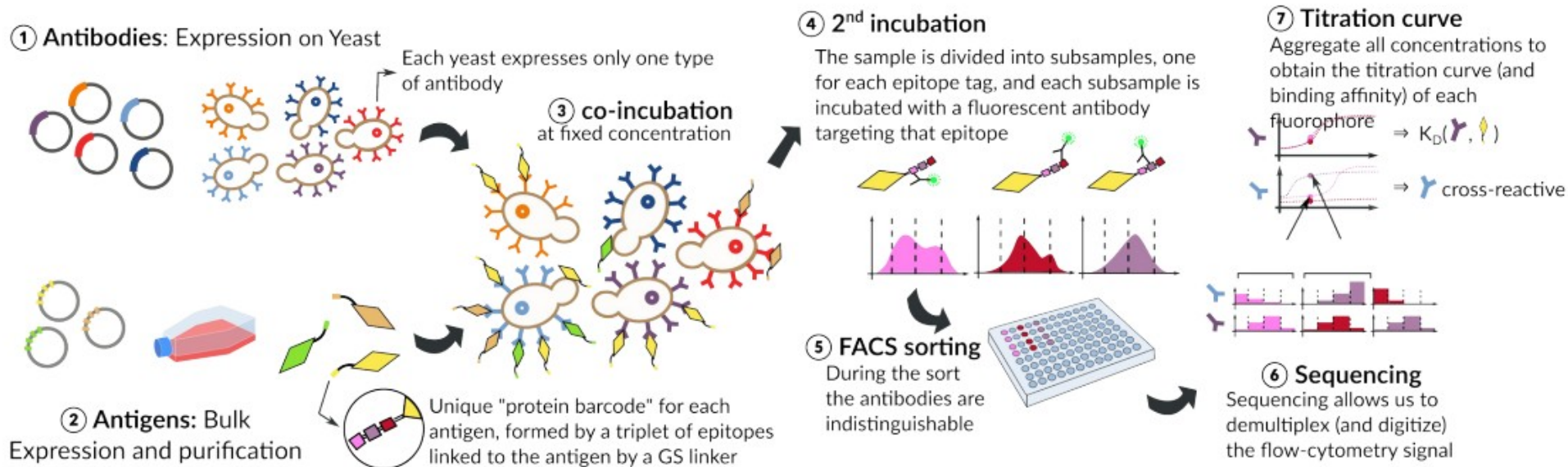


Versus titrage

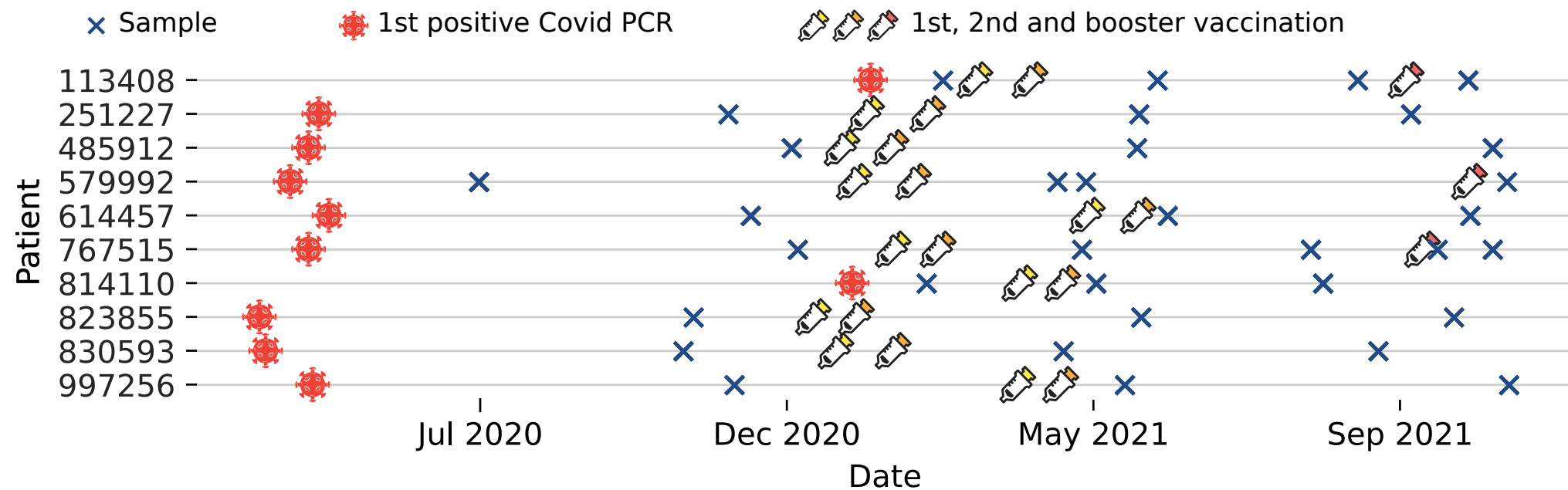


Versus Bio-Layer
interferometry

Tite-Seq ++



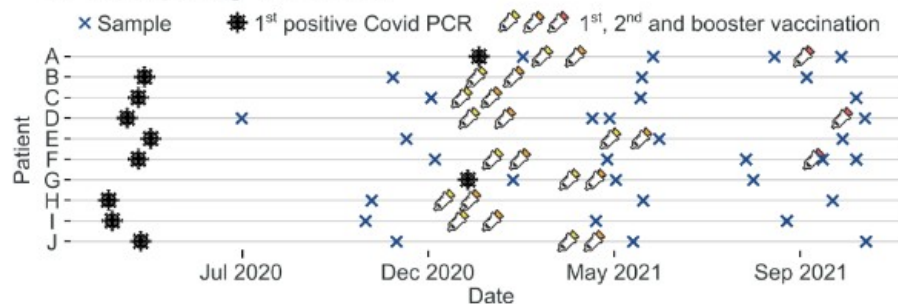
Current project, Covid vaccination



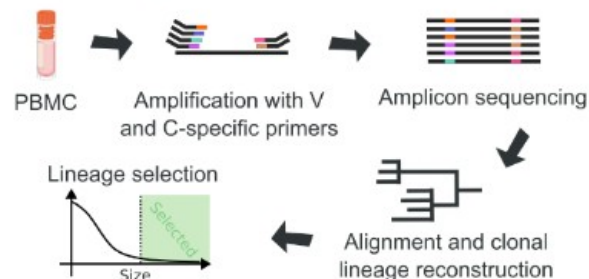
10^9 B-cells.

Current project, Covid vaccination

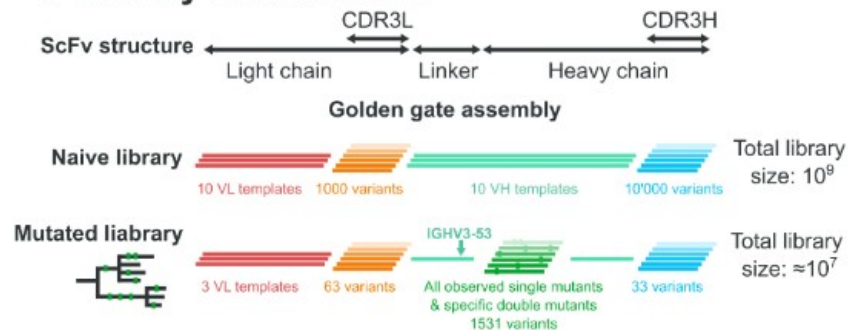
A ① Sampling timeline



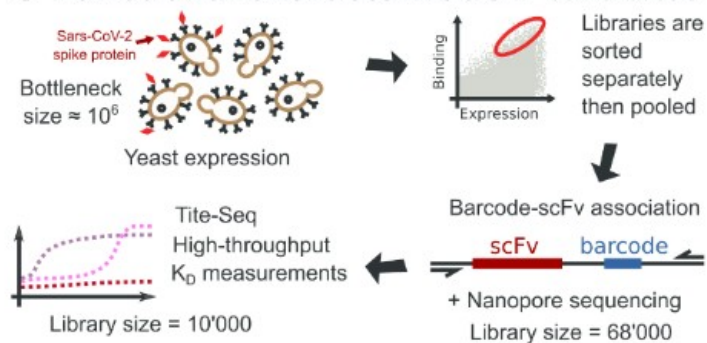
② Sequencing & Repertoire analysis



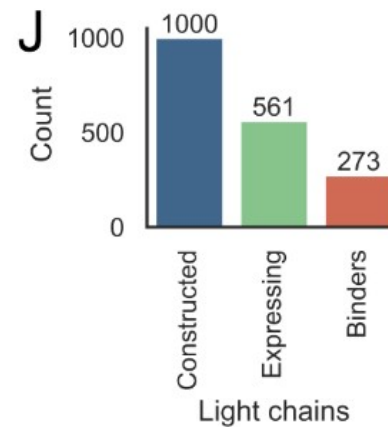
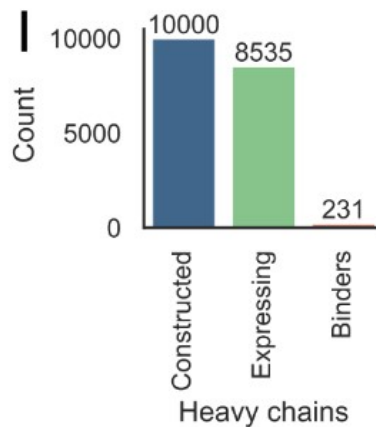
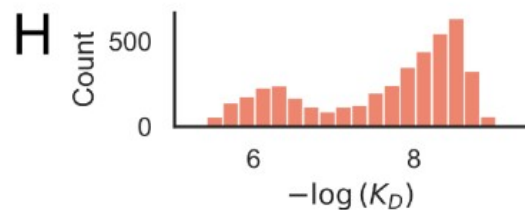
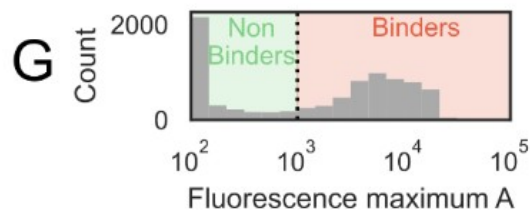
③ Library construction



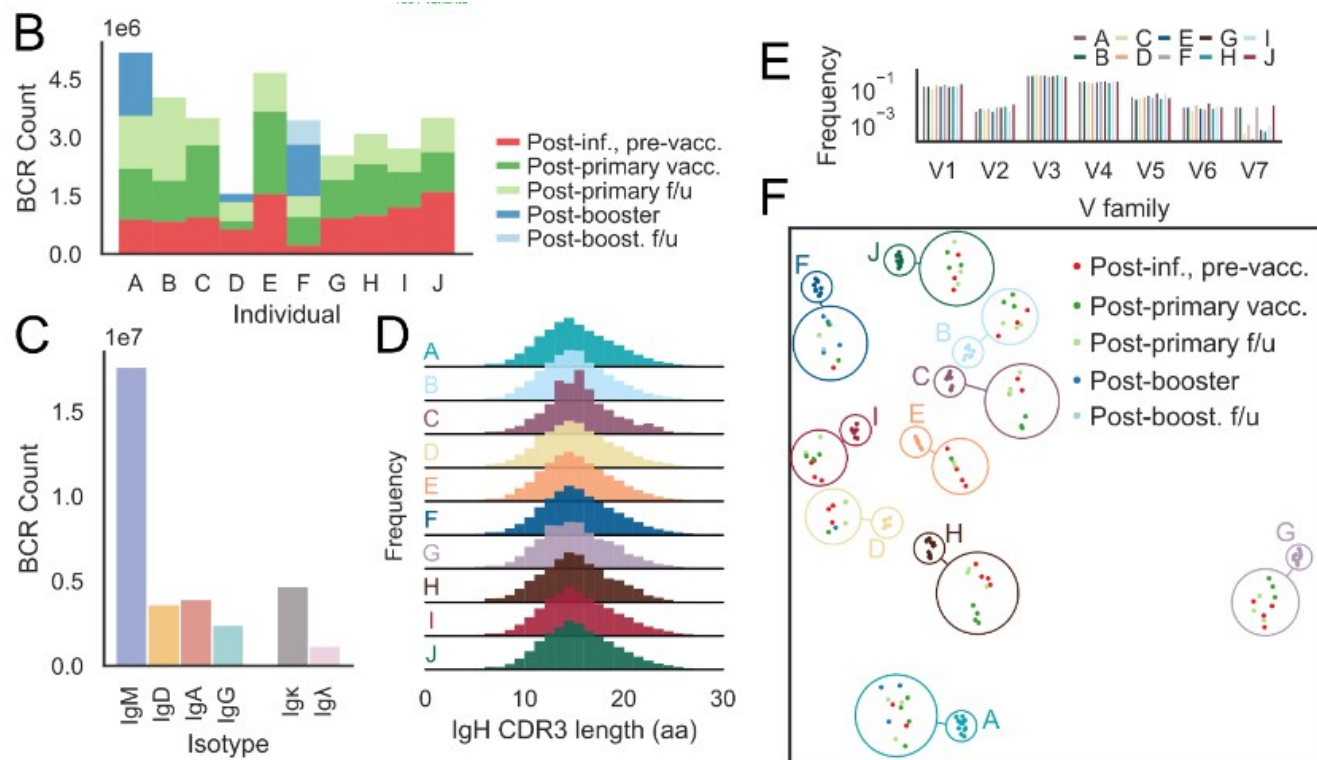
④ Functional characterization



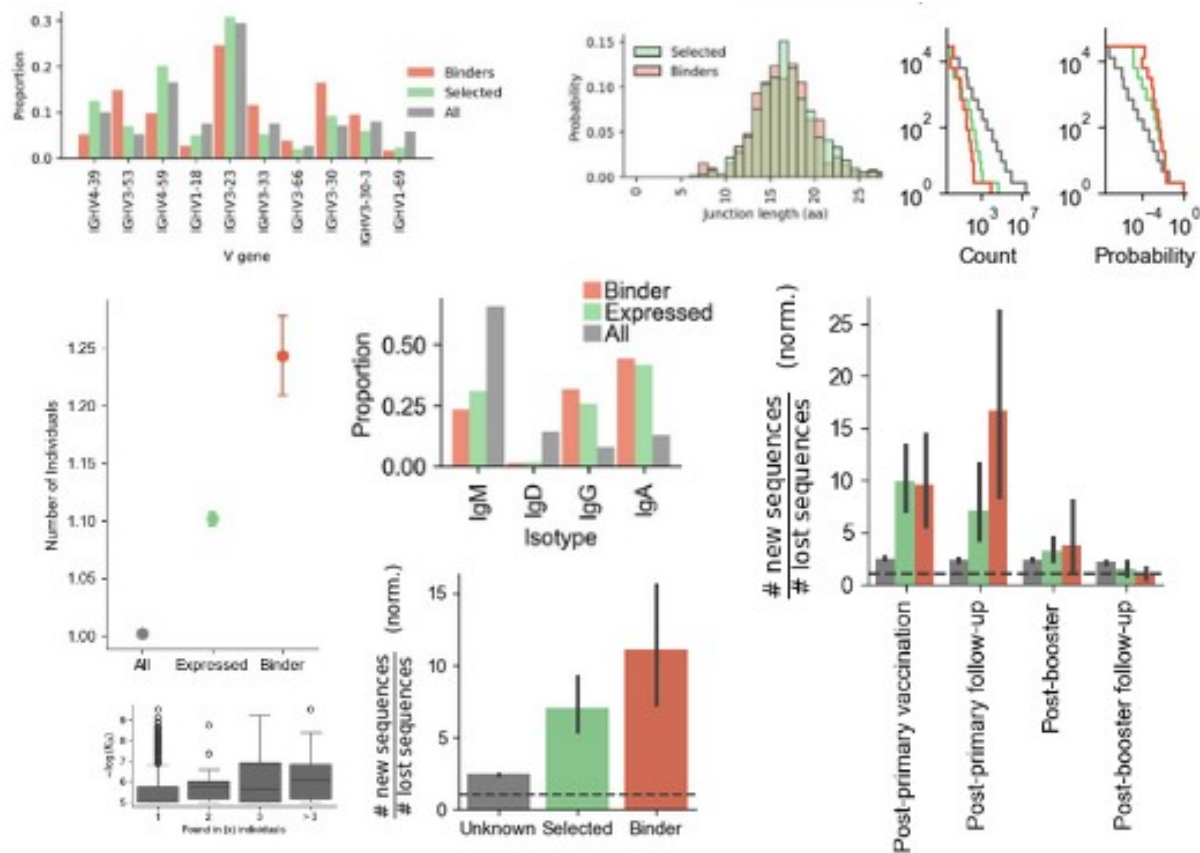
Current project, Covid vaccination



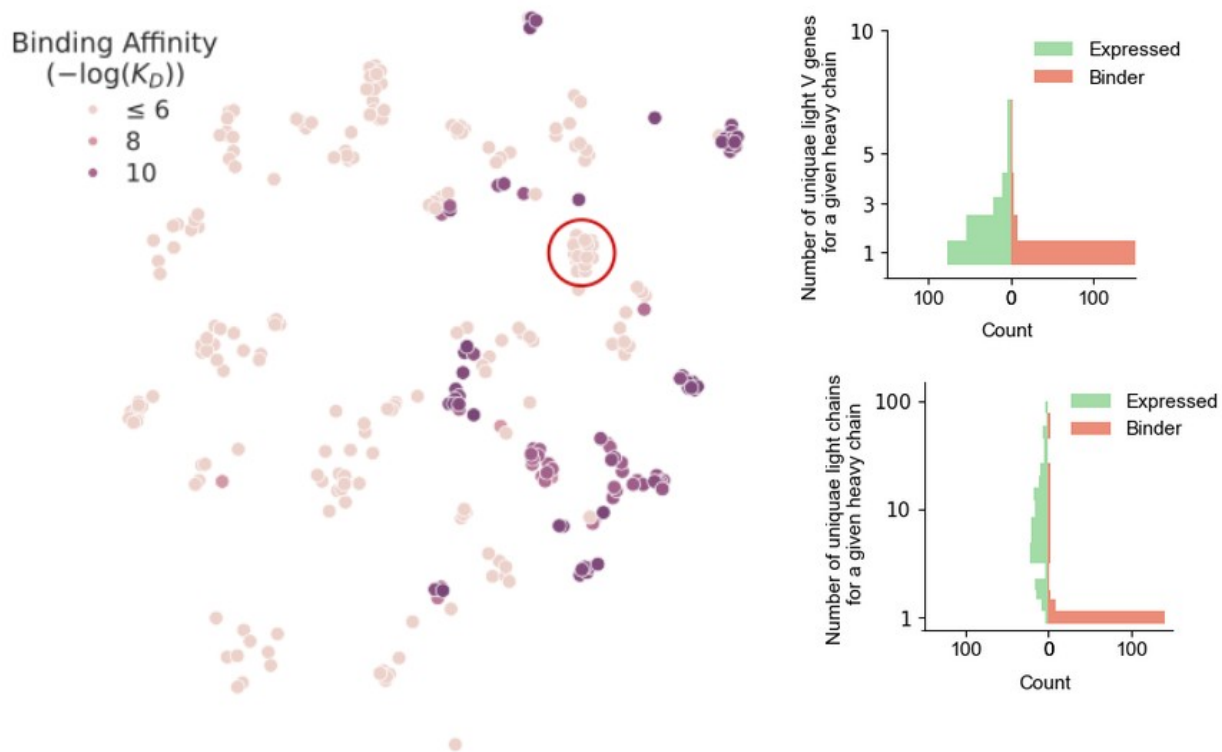
Current project, Covid vaccination



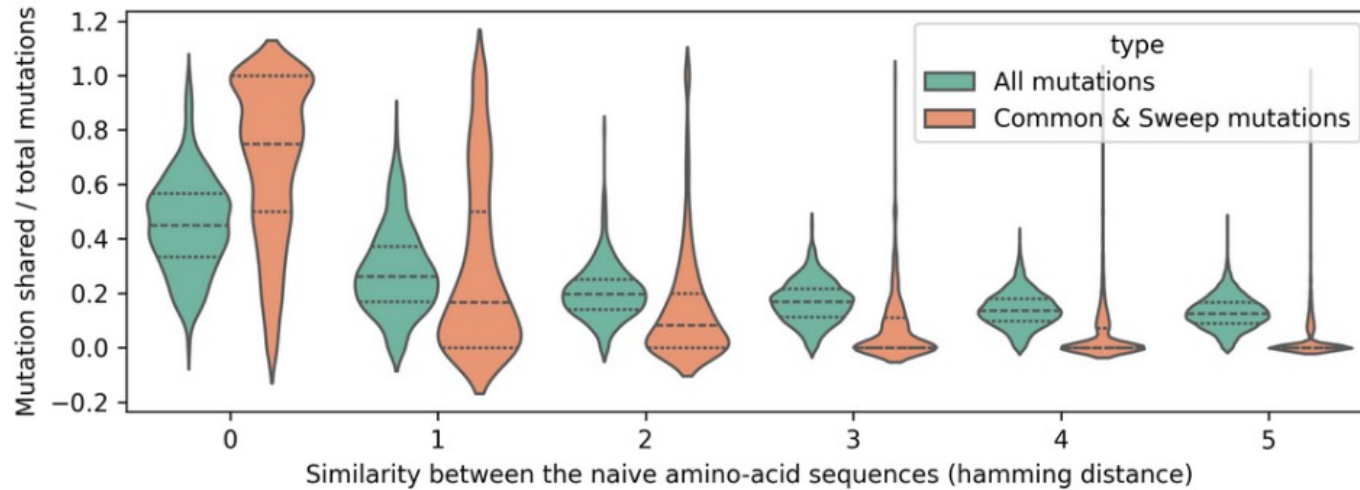
Current project, Covid vaccination



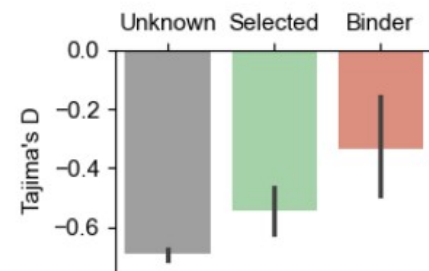
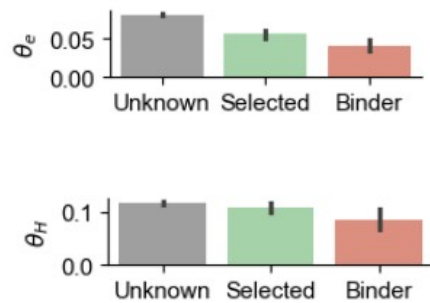
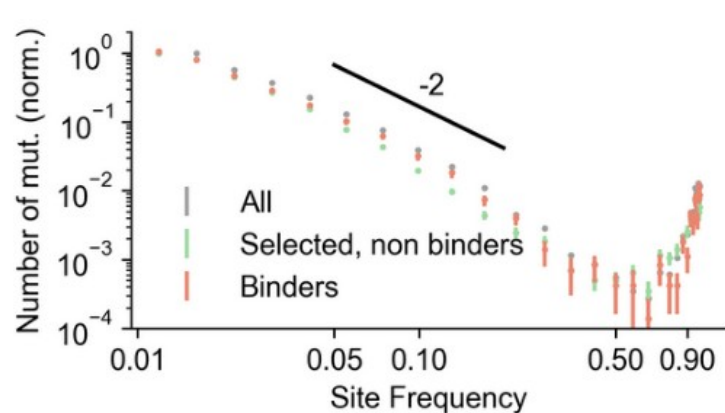
Current project, Covid vaccination



Current project, Covid vaccination



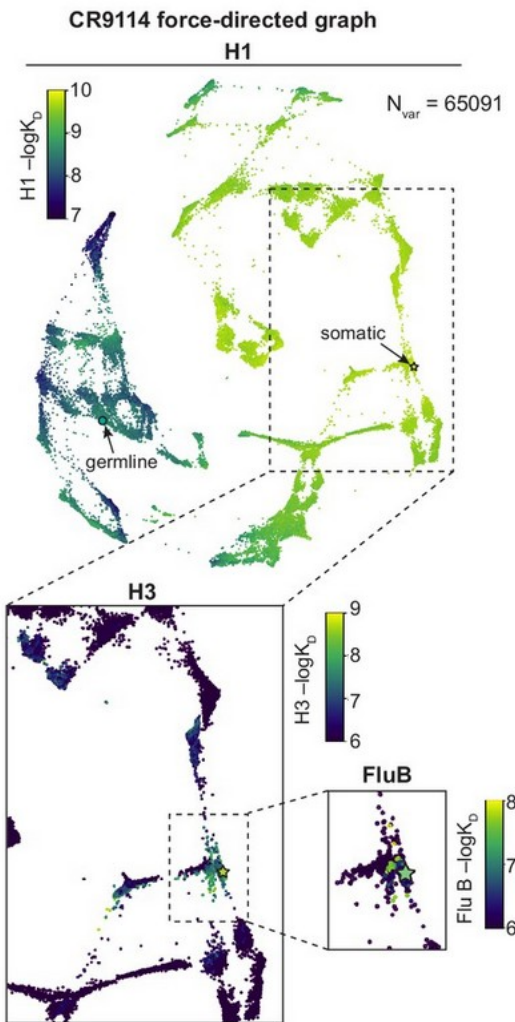
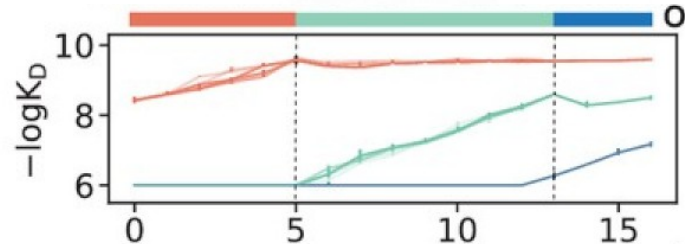
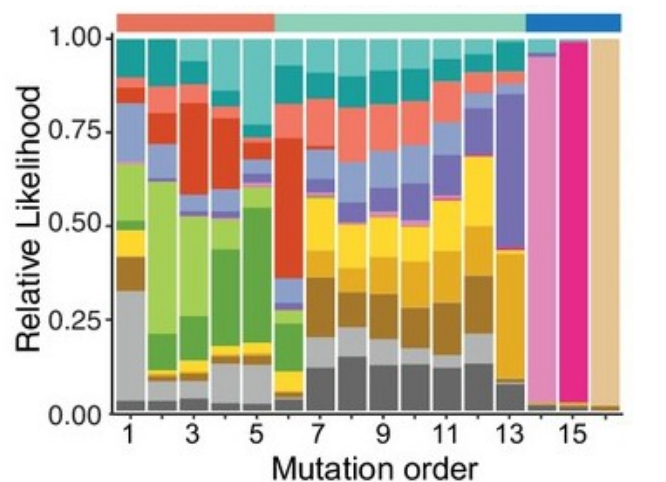
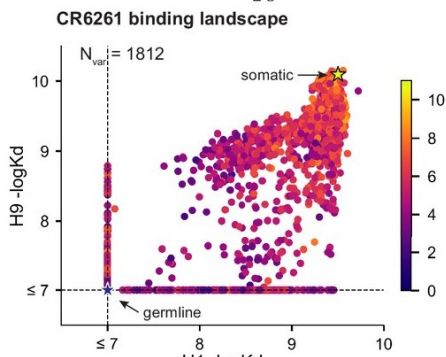
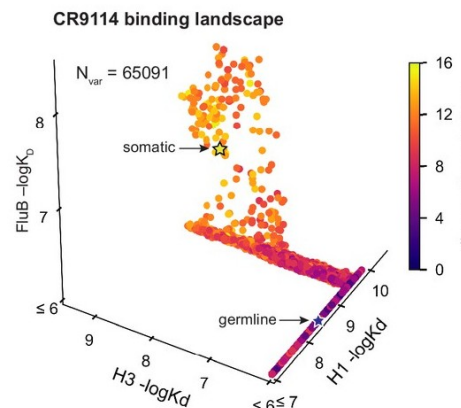
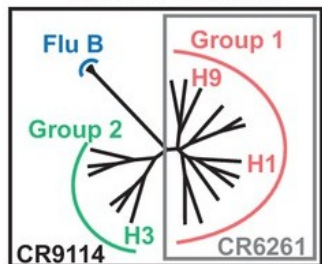
Current project, Covid vaccination



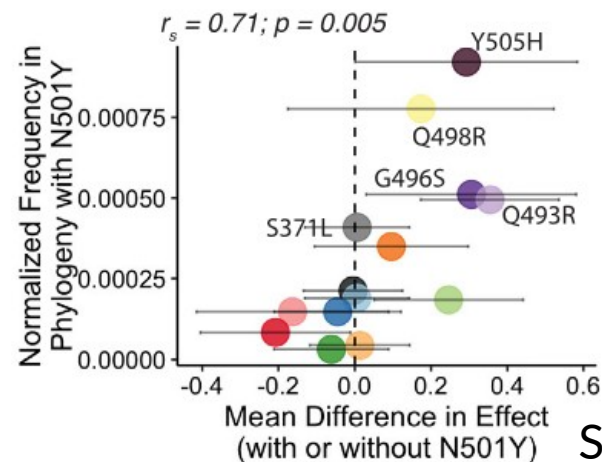
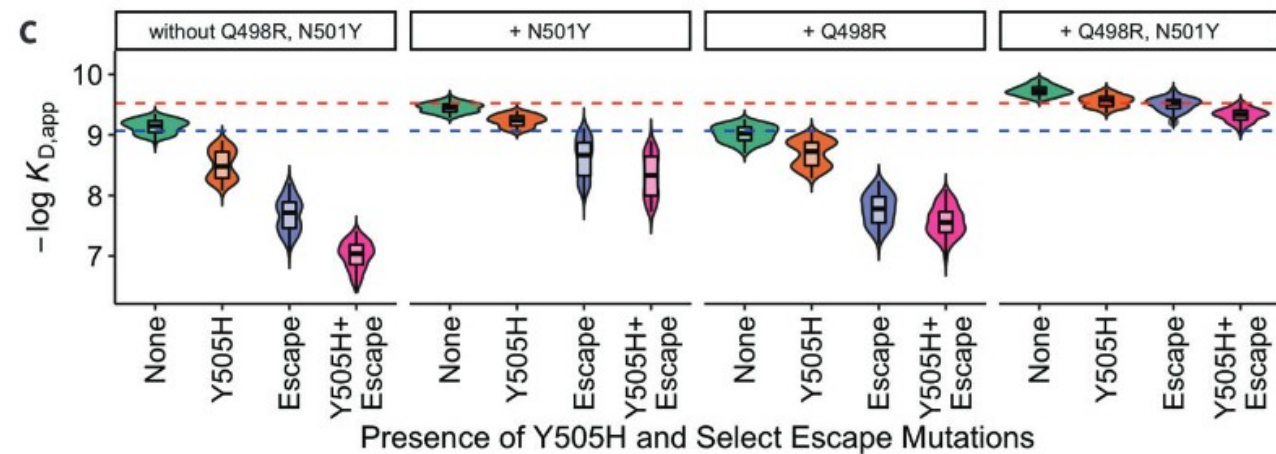
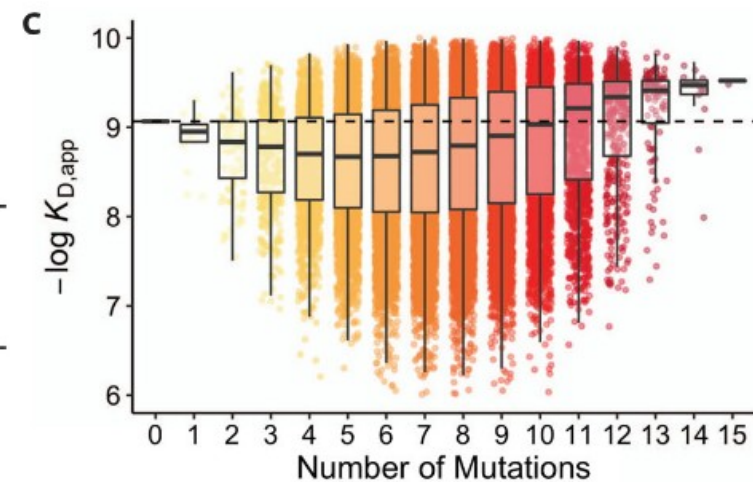
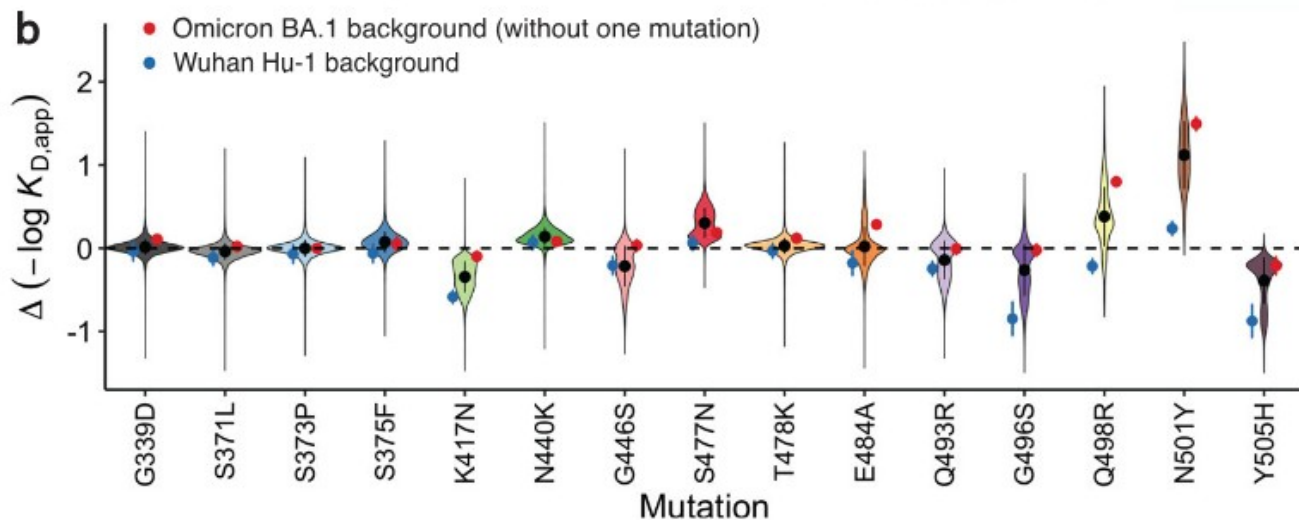
Evolution of a broadly neutralising Ab

	FW1	HCDR1	FW2	HCDR2	FW3	HCDR3	FW4
CR9114 germline	QVQLVQSGAEVKKPGSSVKVCSKASGGTFFSSYAISWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADKSTSTAYMELSSLRSED	TAVYYCARHGNYYYYYYGM	DMVWGQGT	TVTSS			
CR9114 somatic	S.....SNN.....D.....S.....STA.....S.....IFS.....N.....T.....F.....S.....						
		30 35 36		57 64-66	79 82-85	92 95 103	113
CR6261 germline	QVQLVQSGAEVKKPGSSVKVCSKASGGTFFSSYAISWVRQAPGQGLEWMGGIIPIFGTANYAQKFQGRVTITADKSTSTAYMELSSLRSED	TAVYYCAKHMGYQLRETMDVWGQGT	TVTSS				
CR6261 somatic	E.....P.....TK.....P.....DFAG.....V.....M.....V.....						
		29 35		65 66 69	82-85 87		112.1

Influenza HA diversity & bnAb breadth



Evolution of the Omicron variant



Evolution of the Omicron variant

